

Supplementary Material: Provable Tensor Robust Principal Component Analysis via Adaptive Convex-Concave Regularization

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Abstract

This supplementary material provides the detailed proofs and additional experiments supporting the revised manuscript. We first introduce the reduced-core lifting and support operators, the random corruption model, and the modified tensor incoherence condition, and then restate the approximate local recovery theorem. In the recovery proof, the column-orthogonal representation matrix is conditioned on so that the tangent space, support operators, and dual certificate are analyzed in a common coordinate system; the practical algorithm still learns this matrix jointly with the core and sparse variables. The recovery proof is organized as one continuous primal–dual witness chain: a deterministic certificate sufficiency argument, a weighted golfing construction for the low-rank certificate, a Neumann-series correction for the sparse-sign component, and a final high-probability verification. We subsequently prove multiplier boundedness and primal-residual convergence under the uncapped theoretical penalty schedule and show, under the stated boundedness and asymptotic-regularity assumptions, that every accumulation point satisfies generalized Karush–Kuhn–Tucker conditions. Additional real-data and phase-transition experiments are included at the end.

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Appendix A. Main Preliminaries and Problem Formulation

This appendix introduces the notation and operators needed for the recovery analysis and restates the approximate local recovery theorem proved in Appendices B–E. We first distinguish the two tensor unfoldings, define the reduced-core lifting and support operators, and specify the random corruption model and local-solution notion. We then state the modified tensor incoherence condition and the recovery theorem in the same form as in the main manuscript. The detailed proof starts with the deterministic reduction in Appendix B and ends with the final certificate verification in Appendix E. Standard tensor algebra used below can be found in [1].

For a third-order tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, its (i, j, k) -th entry is denoted by x_{ijk} , and its k -th frontal slice is denoted by $\mathcal{X}^{(k)} = \mathcal{X}(:, :, k) \in \mathbb{R}^{n_1 \times n_2}$. The lateral slice and tube are denoted by $\mathcal{X}(:, j, :)$ and $\mathcal{X}(i, j, :)$, respectively.

Definition 1 (Frontal-slice unfolding, [1]). *For $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, define*

$$\text{unfold}(\mathcal{X}) := \begin{bmatrix} \mathcal{X}^{(1)} \\ \mathcal{X}^{(2)} \\ \vdots \\ \mathcal{X}^{(n_3)} \end{bmatrix} \in \mathbb{R}^{n_1 n_3 \times n_2}. \quad (\text{A.1})$$

Its inverse is denoted by $\text{fold}(\cdot)$.

Definition 2 (Mode-3 unfolding, [1]). *For $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, the mode-3 unfolding used in the Procrustes update is defined as*

$$\text{unfold}_3(\mathcal{X}) := [\text{vec}(\mathcal{X}^{(1)}) \quad \text{vec}(\mathcal{X}^{(2)}) \quad \cdots \quad \text{vec}(\mathcal{X}^{(n_3)})] \in \mathbb{R}^{(n_1 n_2) \times n_3}, \quad (\text{A.2})$$

where $\text{vec}(\cdot)$ stacks the columns of a matrix. Equivalently,

$$[\text{unfold}_3(\mathcal{X})]_{i+(j-1)n_1, k} = x_{ijk}. \quad (\text{A.3})$$

For every dimensionally compatible matrix A ,

$$\text{unfold}_3(\mathcal{X} \times_3 A) = \text{unfold}_3(\mathcal{X}) A^T. \quad (\text{A.4})$$

The two unfolding operators are distinct and are not used interchangeably.

Definition 3 (Conditioned reduced-core lifting operator). *Let $D_0 \in \mathbb{R}^{n_3 \times r_3}$ be the column-orthogonal representation matrix on which the recovery analysis is conditioned, with $D_0^T D_0 = I_{r_3}$. Define*

$$\mathcal{A}_{D_0}(\mathcal{H}) := \mathcal{H} \times_3 D_0, \quad \mathcal{A}_{D_0}^T(\mathcal{Z}) := \mathcal{Z} \times_3 D_0^T. \quad (\text{A.5})$$

Then

$$\langle \mathcal{A}_{D_0}(\mathcal{H}), \mathcal{Z} \rangle = \langle \mathcal{H}, \mathcal{A}_{D_0}^T(\mathcal{Z}) \rangle, \quad \mathcal{A}_{D_0}^T \mathcal{A}_{D_0} = \mathcal{I}, \quad \|\mathcal{A}_{D_0}(\mathcal{H})\|_F = \|\mathcal{H}\|_F. \quad (\text{A.6})$$

In contrast, $\mathcal{A}_{D_0} \mathcal{A}_{D_0}^T$ is only the orthogonal projection onto the ambient third-mode subspace spanned by D_0 and is not the identity on the whole ambient tensor space when $r_3 < n_3$.

Definition 4 (Sparse support and induced core operator). *Let $\Gamma := \text{supp}(\mathcal{E}_0)$ be the sparse support. The ambient entrywise support projections are denoted by $\mathcal{P}_\Gamma^{\text{orig}}$ and $\mathcal{P}_{\Gamma^c}^{\text{orig}}$. The induced operator on the reduced core is*

$$\mathcal{R}_\Gamma := \mathcal{A}_{D_0}^T \mathcal{P}_\Gamma^{\text{orig}} \mathcal{A}_{D_0}. \quad (\text{A.7})$$

It is a self-adjoint positive-semidefinite contraction, but is not in general an orthogonal projection. We use Ω exclusively for the CCSVS parameter tuple and Γ exclusively for sparse support.

Assumption Appendix A.1 (Random sparse-corruption model). *The support Γ is generated by independent Bernoulli variables with inclusion probability ρ_s . Conditional on Γ , the nonzero signs of $\mathcal{E}_0$ are independent symmetric Rademacher variables. The high-probability statements in Appendices C–E are with respect to this support-and-sign model, conditional on the representation pair $(\mathcal{B}_0, D_0)$.*

Definition 5 (CCSVS parameters and local solution). *Let $\Omega = (\omega, \omega_1, \dots, \omega_r)$, where $\omega > 0$ and $\omega_1 \geq \dots \geq \omega_r \geq 0$, and set $\tilde{\omega}_j := \omega + \omega_j$ and $\tilde{\omega}_{\max} := \max_{1 \leq j \leq r} \tilde{\omega}_j$. For the positive slice weights, write $\vartheta_{\min} := \min_k \vartheta_k$ and $\vartheta_{\max} := \max_k \vartheta_k$ and impose the standing balance condition $\vartheta_{\min} \geq \vartheta_{\max}/2$. A feasible pair $(\mathcal{B}_0, \mathcal{E}_0)$ is an (ϵ, δ) -asymptotically unique local solution if every nonzero feasible core perturbation $\mathcal{H}$ satisfying $\|\mathcal{H}\|_F \leq \epsilon$ obeys*

$$\begin{aligned} \text{Sep}(\mathcal{B}_0 + \mathcal{H}, \Omega, \vartheta) + \lambda \|\mathcal{E}_0 - \mathcal{A}_{D_0}(\mathcal{H})\|_1 \\ > \text{Sep}(\mathcal{B}_0, \Omega, \vartheta) + \lambda \|\mathcal{E}_0\|_1 - \delta. \end{aligned} \quad (\text{A.8})$$

Throughout the recovery analysis, r_3 is the prescribed reduced third-mode dimension, r is the singular-value separation rank, and R is the tubal rank of the reduced core. The representation matrix D_0 is conditioned on when defining the associated tangent space, support operators, and certificate components. This conditioning is a proof device and does not change the joint optimization of $(\mathcal{B}, D, \mathcal{E})$ in the practical algorithm.

Throughout the recovery analysis, $\omega > 0$ and $\kappa_{\vartheta} := \vartheta_{\max}/\vartheta_{\min} \leq 2$ are treated as fixed numerical constants independent of the tensor dimensions, rank, and corruption probability. Their dependence is therefore absorbed into the generic constants appearing in the concentration bounds.

Building upon these basic notations, we now introduce the *Modified Adjusted Tensor Incoherence Condition*, which forms the fundamental assumption for our recovery guarantees.

Definition 6 (Modified Adjusted Tensor Incoherence Condition). *Let $\mathcal{T}_0 = \mathcal{B}_0 \times_3 D_0$ admit a column-orthogonal reduced representation, where $\mathcal{B}_0 \in \mathbb{R}^{n_1 \times n_2 \times r_3}$, $D_0 \in \mathbb{R}^{n_3 \times r_3}$, and $D_0^T D_0 = I_{r_3}$. Here, r_3 is the prescribed reduced third-mode dimension, i.e., the number of columns of D_0 , with $1 \leq r_3 < n_3$.*

Throughout this definition, $$ denotes the standard Fourier-based t -product on the reduced core space along its third mode of size r_3 . Let*

$$\mathcal{B}_0 = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$$

be the skinny t -SVD of $\mathcal{B}_0$, with tubal rank R .

We say that $\mathcal{T}_0$ satisfies the adjusted tensor incoherence condition with core incoherence parameter $\mu \geq 1$ and transform row-incoherence parameter $\mu_D \geq 1$ if the following conditions hold.

1. Core Incoherence:

$$\begin{aligned} \max_{i=1, \dots, n_1} \left\| \mathcal{U}^T * e_i^{(1)} \right\|_F &\leq \sqrt{\frac{\mu R}{n_1}}, \\ \max_{j=1, \dots, n_2} \left\| \mathcal{V}^T * e_j^{(2)} \right\|_F &\leq \sqrt{\frac{\mu R}{n_2}}, \\ \left\| \mathcal{U} * \mathcal{V}^T \right\|_\infty &\leq \sqrt{\frac{\mu R}{n_1 n_2}}. \end{aligned} \tag{A.9}$$

Here, $\{e_i^{(1)}\}_{i=1}^{n_1}$ and $\{e_j^{(2)}\}_{j=1}^{n_2}$ denote the standard tensor column bases induced by the Euclidean bases of $\mathbb{R}^{n_1}$ and $\mathbb{R}^{n_2}$, respectively; specifically,

$e_i^{(1)} \in \mathbb{R}^{n_1 \times 1 \times r_3}$ has one at $(i, 1, 1)$ and zeros elsewhere, and $e_j^{(2)} \in \mathbb{R}^{n_2 \times 1 \times r_3}$ is defined analogously.

2. **Transform Row Incoherence:** We say that the column-orthogonal transform $D_0 \in \mathbb{R}^{n_3 \times r_3}$ is row-incoherent with parameter $\mu_D \geq 1$ if

$$\max_{k=1, \dots, n_3} \|D_0(k, :)\|_2^2 \leq \frac{\mu_D r_3}{n_3}, \quad (\text{A.10})$$

where $D_0(k, :)$ denotes the k -th row of D_0 . The smallest admissible row-incoherence parameter is

$$\mu_D(D_0) := \frac{n_3}{r_3} \max_{k=1, \dots, n_3} \|D_0(k, :)\|_2^2. \quad (\text{A.11})$$

Since $D_0^T D_0 = I_{r_3}$, we have $\sum_{k=1}^{n_3} \|D_0(k, :)\|_2^2 = r_3$, and hence $1 \leq \mu_D(D_0) \leq n_3/r_3$.

Building upon the modified adjusted tensor incoherence condition, we now state the approximate recovery theorem for the reduced-core AC²-TRPCA model. In the proof, the column-orthogonal representation matrix D_0 is conditioned on so that the tangent space, support projections, dual certificate, and random operators are defined with respect to a common coordinate system. This is a technical device used in the recovery analysis and does not alter the joint optimization of $(\mathcal{B}, D, \mathcal{E})$ in the proposed algorithm.

Theorem 1 (Approximate recovery guarantee). *Consider the observation model $\mathcal{Y} = \mathcal{T}_0 + \mathcal{E}_0 \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, where $\mathcal{E}_0$ is a sparse corruption tensor and the low-rank component admits the column-orthogonal reduced representation $\mathcal{T}_0 = \mathcal{B}_0 \times_3 D_0$ with $D_0^T D_0 = I_{r_3}$. Assume that $(\mathcal{B}_0, D_0)$ satisfies the modified adjusted tensor incoherence condition with tubal rank R and parameters (μ, μ_D) .*

For the recovery analysis, condition on D_0 and consider

$$\min_{\mathcal{B}, \mathcal{E}} \text{Sep}(\mathcal{B}, \Omega, \vartheta) + \lambda \|\mathcal{E}\|_1 \quad \text{s.t.} \quad \mathcal{Y} = \mathcal{B} \times_3 D_0 + \mathcal{E}. \quad (\text{A.12})$$

Assume additionally that the sparse corruption follows Assumption [Appendix A.1](#). Suppose that there exists a universal constant $c_0 > 0$ such that:

1. *The regularization parameter is $\lambda = 1/\sqrt{n_{(1)}}$, where $n_{(1)} = \max\{n_1, n_2\}$.*

2. The tubal rank satisfies $R \leq c_0 \mu^{-1} n_{(2)} \log^{-2}(n_{(1)})$, where $n_{(2)} = \min\{n_1, n_2\}$.
3. The corruption probability satisfies $\rho_s \leq c_0/(\mu_D r_3)$.

For the prescribed separation parameters, define $\vartheta_{\max} := \max_{1 \leq i \leq r_3} \vartheta_i$ and $\tilde{\omega}_{\max} := \max_{1 \leq j \leq r} \tilde{\omega}_j$. Then there exist universal constants $c_1, c_2 > 0$ such that, with probability at least $1 - c_1(n_{(1)} n_3)^{-c_2}$, the ground-truth pair $(\mathcal{B}_0, \mathcal{E}_0)$ is an (ϵ, δ) -asymptotically unique local solution of problem (A.12), where

$$\delta := \epsilon \left(\frac{\lambda \sqrt{r_3}}{4} + \vartheta_{\max} \tilde{\omega}_{\max} r r_3 \right). \quad (\text{A.13})$$

Remark 1 (Role of the conditioned representation matrix). In Algorithm 1, the column-orthogonal matrix D is learned from the observed data together with $\mathcal{B}$ and $\mathcal{E}$. In Theorem 1, the representation matrix D_0 is conditioned on only for the purpose of the recovery proof. Holding it fixed allows all tangent-space projections, support operators, and certificate components to be analyzed in a common coordinate system.

This treatment is compatible with adaptive column-orthogonal learning. Indeed, different orthonormal bases of the same reduced subspace only change the coordinates of the core tensor and leave the represented full tensor unchanged, as formalized in Corollary 1.

Corollary 1 (Representation invariance under an orthogonal basis change). Let $D \in \mathbb{R}^{n_3 \times r_3}$ satisfy $D^T D = I_{r_3}$ and $\text{range}(D) = \text{range}(D_0)$. Define $Q := D^T D_0$ and $\mathcal{B}_{0,D} := \mathcal{B}_0 \times_3 Q$. Then Q is orthogonal and $\mathcal{T}_0 = \mathcal{B}_{0,D} \times_3 D$.

If $(\mathcal{B}_{0,D}, D)$ satisfies the same incoherence condition and the three sufficient conditions of Theorem 1, then the same approximate local recovery conclusion holds when the recovery analysis is conditioned on D in place of D_0 .

Proof. Because D and D_0 have orthonormal columns and the same range, $DD^T = D_0 D_0^T$. Hence $Q^T Q = D_0^T DD^T D_0 = I_{r_3}$, so Q is orthogonal, and $DQ = DD^T D_0 = D_0$. Therefore,

$$\mathcal{B}_{0,D} \times_3 D = (\mathcal{B}_0 \times_3 Q) \times_3 D = \mathcal{B}_0 \times_3 (DQ) = \mathcal{T}_0.$$

The conclusion follows by applying Theorem 1 to the reparameterized pair $(\mathcal{B}_{0,D}, D)$ under the stated conditions. $\square$

The following auxiliary estimate is not needed for the proof of Theorem 1; it is included only to record the deterministic projection error associated with another column-orthogonal subspace.

Proposition 1 (Auxiliary projection error bound). *Let $D^T D = I_{r_3}$, $P_D := DD^T$, $\mathcal{B}_{0,D} := \mathcal{T}_0 \times_3 D^T$, and $\eta_D := \|(I - P_D)D_0\|_2$. Then*

$$\mathcal{B}_{0,D} \times_3 D = \mathcal{T}_0 \times_3 P_D, \quad \|\mathcal{T}_0 - \mathcal{B}_{0,D} \times_3 D\|_F \leq \eta_D \|\mathcal{B}_0\|_F. \quad (\text{A.14})$$

For every estimated core $\widehat{\mathcal{B}}$,

$$\|\widehat{\mathcal{B}} \times_3 D - \mathcal{T}_0\|_F \leq \|\widehat{\mathcal{B}} - \mathcal{B}_{0,D}\|_F + \eta_D \|\mathcal{B}_0\|_F. \quad (\text{A.15})$$

Proof. Associativity gives $\mathcal{B}_{0,D} \times_3 D = \mathcal{T}_0 \times_3 DD^T$. Moreover, $\mathcal{T}_0 - \mathcal{T}_0 \times_3 P_D = \mathcal{B}_0 \times_3 [(I - P_D)D_0]$. Using $\|\mathcal{X} \times_3 A\|_F \leq \|A\|_2 \|\mathcal{X}\|_F$ proves the first bound. The second follows from the triangle inequality and the Frobenius isometry of multiplication by the column-orthogonal matrix D . $\square$

Proof roadmap for Theorem 1. The recovery proof follows a primal–dual witness architecture, but each appendix has a distinct role. Appendix B gives the deterministic implication: if a certificate satisfies the tangent-space, support, and norm conditions, then every nonzero feasible perturbation has objective gap greater than $-\delta$. This is also where the concave CCSVS term is controlled locally. Appendix C constructs the low-rank certificate component by a weighted golfing recursion. Its projection-energy, unbiasedness, and concentration lemmas are needed to prove geometric contraction and the required spectral and entrywise bounds. Appendix D adds the sparse-sign correction through an invertible complement operator and a Neumann series; the associated lemmas establish restricted identifiability, the exact cancellation identity, and the remaining norm bounds. Appendix E combines the two components and verifies all hypotheses of the deterministic lemma on one high-probability event. The proof of Theorem 1 is completed explicitly at the end of Appendix E.

The tangent-space decomposition, golfing construction, Neumann-series correction, and concentration tools are standard ingredients in low-rank-plus-sparse recovery [2, 11]. The adaptations here are the reduced-core operators induced by the conditioned representation matrix D_0 , the dependence on (μ_D, r_3) , the weighted slice operator, and the local bound for the concave CCSVS compensation term.

Appendix B. Sufficiency of Dual Certificate

This appendix begins the proof of Theorem 1. Its purpose is purely deterministic: it identifies certificate conditions sufficient for local recovery, without yet addressing whether such a certificate exists with high probability. Lemma 1 is therefore the bridge between certificate construction and the theorem. Its proof controls the convex spectral part, uses the certificate identity to cancel the sparse term, bounds the concave compensation, and invokes restricted identifiability to obtain strictness. Here $\Gamma = \text{supp}(\mathcal{E}_0)$ is the sparse support, whereas Ω remains the CCSVS parameter tuple.

Lemma 1 (Sufficiency of Dual Certificate). *Let $\mathcal{B}_0 = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$ be the skinny reduced-core t -SVD with tubal rank R , and set $\mathcal{S}_0 := \mathcal{U} * \mathcal{V}^T$. Define*

$$\mathcal{T}_U := \{\mathcal{U} * \mathcal{Y} : \mathcal{Y} \in \mathbb{R}^{R \times n_2 \times r_3}\}, \quad (\text{B.1})$$

$$\mathcal{T}_V := \{\mathcal{X} * \mathcal{V}^T : \mathcal{X} \in \mathbb{R}^{n_1 \times R \times r_3}\}, \quad \mathcal{T} := \mathcal{T}_U + \mathcal{T}_V. \quad (\text{B.2})$$

Assume the restricted-identifiability condition

$$\mathcal{H} \in \mathcal{T}, \quad \mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{A}_{D_0}(\mathcal{H}) = 0 \implies \mathcal{H} = 0. \quad (\text{B.3})$$

Suppose there exist $(\mathcal{W}, \mathcal{F}, \mathcal{D})$ satisfying

$$\omega(\vartheta \otimes (\mathcal{S}_0 + \mathcal{W})) = \lambda \mathcal{A}_{D_0}^T \left(\text{sgn}(\mathcal{E}_0) + \mathcal{F} + \mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{D} \right), \quad (\text{B.4})$$

and

$$\mathcal{P}_{\mathcal{T}} \mathcal{W} = 0, \quad \|\mathcal{W}\| \leq \frac{1}{4}, \quad \mathcal{P}_{\Gamma}^{\text{orig}} \mathcal{F} = 0, \quad \|\mathcal{F}\|_{\infty} \leq \frac{1}{2}, \quad \|\mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{D}\|_F \leq \frac{\sqrt{r_3}}{4}. \quad (\text{B.5})$$

Then every nonzero feasible core perturbation $\mathcal{H}$ with $\|\mathcal{H}\|_F \leq \epsilon$ satisfies

$$\begin{aligned} \Delta(\mathcal{H}) &\geq \frac{\omega \vartheta_{\min}}{2} \|\mathcal{P}_{\mathcal{T}^\perp} \mathcal{H}\|_* + \frac{\lambda}{2} \|\mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{A}_{D_0}(\mathcal{H})\|_1 \\ &\quad - \left(\frac{\lambda \sqrt{r_3}}{4} + \vartheta_{\max} \tilde{\omega}_{\max} r r_3 \right) \|\mathcal{H}\|_F, \end{aligned} \quad (\text{B.6})$$

where

$$\begin{aligned} \Delta(\mathcal{H}) &:= \text{Sep}(\mathcal{B}_0 + \mathcal{H}, \Omega, \vartheta) - \text{Sep}(\mathcal{B}_0, \Omega, \vartheta) \\ &\quad + \lambda (\|\mathcal{E}_0 - \mathcal{A}_{D_0}(\mathcal{H})\|_1 - \|\mathcal{E}_0\|_1). \end{aligned} \quad (\text{B.7})$$

Consequently, $(\mathcal{B}_0, \mathcal{E}_0)$ is an (ϵ, δ) -asymptotically unique local solution with

$$\delta = \epsilon \left(\frac{\lambda\sqrt{r_3}}{4} + \vartheta_{\max} \tilde{\omega}_{\max} r r_3 \right).$$

Proof. Write $\text{Sep} = \Phi - \Psi$, where

$$\Phi(\mathcal{B}) := \omega \sum_{k=1}^{r_3} \vartheta_k \|\widehat{\mathcal{B}}^{(k)}\|_*, \quad (\text{B.8})$$

$$\Psi(\mathcal{B}) := \sum_{k=1}^{r_3} \vartheta_k \sum_{j=1}^r \tilde{\omega}_j \sigma_j(\widehat{\mathcal{B}}^{(k)}). \quad (\text{B.9})$$

Step 1: Convex spectral part. By decomposability of the weighted tensor nuclear norm, and using $\mathcal{P}_{\mathcal{T}}\mathcal{W} = 0$, $\|\mathcal{W}\| \leq 1/4$, and $\vartheta_{\min} \geq \vartheta_{\max}/2$,

$$\Phi(\mathcal{B}_0 + \mathcal{H}) - \Phi(\mathcal{B}_0) \geq \langle \omega(\vartheta \otimes (\mathcal{S}_0 + \mathcal{W})), \mathcal{H} \rangle + \frac{\omega\vartheta_{\min}}{2} \|\mathcal{P}_{\mathcal{T}^\perp} \mathcal{H}\|_*. \quad (\text{B.10})$$

Step 2: Sparse term and dual cancellation. Set $\mathcal{G} := \mathcal{A}_{D_0}(\mathcal{H})$. The support-wise ℓ_1 inequality gives

$$\|\mathcal{E}_0 - \mathcal{G}\|_1 - \|\mathcal{E}_0\|_1 \geq -\langle \text{sgn}(\mathcal{E}_0), \mathcal{G} \rangle + \|\mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{G}\|_1. \quad (\text{B.11})$$

Taking the inner product of (B.4) with $\mathcal{H}$ and using adjointness gives

$$\begin{aligned} & \langle \omega(\vartheta \otimes (\mathcal{S}_0 + \mathcal{W})), \mathcal{H} \rangle \\ &= \lambda \left\langle \text{sgn}(\mathcal{E}_0) + \mathcal{F} + \mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{D}, \mathcal{G} \right\rangle. \end{aligned} \quad (\text{B.12})$$

Since $\mathcal{P}_{\Gamma}^{\text{orig}} \mathcal{F} = 0$ and $\|\mathcal{F}\|_\infty \leq 1/2$,

$$|\langle \mathcal{F}, \mathcal{G} \rangle| \leq \frac{1}{2} \|\mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{G}\|_1.$$

By Cauchy–Schwarz and the lifting isometry,

$$|\langle \mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{D}, \mathcal{G} \rangle| \leq \frac{\sqrt{r_3}}{4} \|\mathcal{H}\|_F. \quad (\text{B.13})$$

Combining these relations with (B.10) gives

$$\begin{aligned} & \Phi(\mathcal{B}_0 + \mathcal{H}) - \Phi(\mathcal{B}_0) + \lambda(\|\mathcal{E}_0 - \mathcal{G}\|_1 - \|\mathcal{E}_0\|_1) \\ & \geq \frac{\omega\vartheta_{\min}}{2} \|\mathcal{P}_{\mathcal{T}^\perp} \mathcal{H}\|_* + \frac{\lambda}{2} \|\mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{G}\|_1 - \frac{\lambda\sqrt{r_3}}{4} \|\mathcal{H}\|_F. \end{aligned} \quad (\text{B.14})$$

Step 3: Concave compensation term. Ky Fan’s inequality, Cauchy–Schwarz, and Parseval’s identity imply

$$\Psi(\mathcal{B}_0 + \mathcal{H}) - \Psi(\mathcal{B}_0) \leq \vartheta_{\max} \tilde{\omega}_{\max} \sqrt{r} \sum_{k=1}^{r_3} \|\widehat{\mathcal{H}}^{(k)}\|_F \quad (\text{B.15})$$

$$\leq \vartheta_{\max} \tilde{\omega}_{\max} r r_3 \|\mathcal{H}\|_F. \quad (\text{B.16})$$

The final, slightly loose form is retained to agree with the theorem’s stated tolerance.

Step 4: Objective gap and strictness. Subtracting (B.16) from Eq. (B.14) proves Eq. (B.6). For $\|\mathcal{H}\|_F \leq \epsilon$, its negative part is bounded by $-\delta$. If both positive terms vanished, then $\mathcal{H} \in \mathcal{T}$ and $\mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{A}_{D_0}(\mathcal{H}) = 0$, so restricted identifiability would imply $\mathcal{H} = 0$. Thus every nonzero feasible perturbation gives a strict gap larger than $-\delta$. $\square$

Lemma 1 completes the deterministic half of the recovery argument. It remains to construct a certificate under the random corruption model. Appendix C produces the low-rank component and a small terminal residual; Appendix D uses the sparse signs and a Neumann-series correction to remove that residual.

Appendix C. Construction of the Dual Certificate via Golfing Scheme

This appendix constructs the low-rank certificate component $(\mathcal{W}^L, \mathcal{F}^L)$ required by Lemma 1. It must keep $\mathcal{W}^L$ in $\mathcal{T}^\perp$, keep $\mathcal{F}^L$ outside the corruption support, and reduce the remaining tangent residual to a negligible size. We use a weighted golfing recursion. The scheme follows standard low-rank tensor recovery arguments [2, 11], while sampling acts in the ambient tensor space and the certificate is represented in the reduced core. Thus the ambient projections $\mathcal{P}_{\Gamma}^{\text{orig}}$ and $\mathcal{P}_{\Gamma^c}^{\text{orig}}$ remain distinct from the induced core operator $\mathcal{R}_{\Gamma}$ in Definition 4.

For positive slice weights, define the linear weighting operator $\mathcal{L}_{\vartheta}$ on the reduced core by

$$\widehat{(\mathcal{L}_{\vartheta} \mathcal{X})}^{(k)} := \vartheta_k \widehat{\mathcal{X}}^{(k)}, \quad k = 1, \dots, r_3. \quad (\text{C.1})$$

Its inverse is obtained by replacing ϑ_k with ϑ_k^{-1} . Since the tangent-space projection is applied slice-wise in the Fourier domain,

$$\mathcal{P}_{\mathcal{T}}\mathcal{L}_{\vartheta} = \mathcal{L}_{\vartheta}\mathcal{P}_{\mathcal{T}}, \quad \mathcal{P}_{\mathcal{T}^{\perp}}\mathcal{L}_{\vartheta} = \mathcal{L}_{\vartheta}\mathcal{P}_{\mathcal{T}^{\perp}}. \quad (\text{C.2})$$

Moreover, with $\kappa_{\vartheta} := \vartheta_{\max}/\vartheta_{\min} \leq 2$,

$$\vartheta_{\min}\|\mathcal{X}\|_F \leq \|\mathcal{L}_{\vartheta}\mathcal{X}\|_F \leq \vartheta_{\max}\|\mathcal{X}\|_F, \quad \|\mathcal{L}_{\vartheta}^{-1}\mathcal{X}\|_F \leq \vartheta_{\min}^{-1}\|\mathcal{X}\|_F. \quad (\text{C.3})$$

The following lemma states the target of this appendix. The projection and concentration lemmas that follow are introduced specifically to verify its geometric contraction, spectral-norm bound, and off-support infinity-norm bound.

Lemma 2 (Low-rank certificate by the golfing scheme). *Let $\mathcal{S}_0 := \mathcal{U}\mathcal{V}^T$ and assume the modified incoherence conditions. Let $\Gamma \sim \text{Ber}(\rho_s)$ and represent its complement as*

$$\Gamma^c = \bigcup_{j=1}^{j_0} \Gamma_j,$$

where the $\Gamma_j \sim \text{Ber}(q)$ are independent and

$$(1-q)^{j_0} = \rho_s, \quad j_0 := \lceil 10 \log n_{(1)} \rceil. \quad (\text{C.4})$$

Set $\mathcal{Y}_0 = 0$ and $\mathcal{R}_0 = \mathcal{S}_0$. Define

$$\mathcal{Y}_j := \mathcal{Y}_{j-1} + q^{-1}\mathcal{L}_{\vartheta}^{-1}\mathcal{A}_{D_0}^T \mathcal{P}_{\Gamma_j}^{\text{orig}} \mathcal{A}_{D_0} \mathcal{L}_{\vartheta}(\mathcal{R}_{j-1}), \quad (\text{C.5})$$

$$\mathcal{R}_j := \mathcal{P}_{\mathcal{T}}(\mathcal{S}_0 - \mathcal{Y}_j), \quad \mathcal{W}^L := \mathcal{P}_{\mathcal{T}^{\perp}}\mathcal{Y}_{j_0}, \quad (\text{C.6})$$

$$\mathcal{F}^L := \frac{\omega}{\lambda q} \sum_{j=1}^{j_0} \mathcal{P}_{\Gamma_j}^{\text{orig}} \mathcal{A}_{D_0} \mathcal{L}_{\vartheta}(\mathcal{R}_{j-1}). \quad (\text{C.7})$$

If

$$q \geq C_0 \kappa_{\vartheta}^2 \frac{\mu \mu_D R r_3 \log(n_{(1)} n_3)}{n_{(2)} n_3}, \quad (\text{C.8})$$

then, for a sufficiently large universal constant C_0 , with probability at least $1 - C(n_{(1)} n_3)^{-10}$,

$$\mathcal{P}_{\mathcal{T}}\mathcal{W}^L = 0, \quad \|\mathcal{W}^L\| \leq \frac{1}{8}, \quad \mathcal{P}_{\Gamma}^{\text{orig}}\mathcal{F}^L = 0, \quad \|\mathcal{F}^L\|_{\infty} \leq \frac{1}{4}, \quad (\text{C.9})$$

and

$$\|\mathcal{R}_{j_0}\|_F \leq 2^{-j_0} \|\mathcal{S}_0\|_F \leq n_{(1)}^{-9} \|\mathcal{S}_0\|_F. \quad (\text{C.10})$$

In addition, the following exact identity holds:

$$\omega \mathcal{L}_\vartheta(\mathcal{S}_0 + \mathcal{W}^L) = \lambda \mathcal{A}_{D_0}^T(\mathcal{F}^L) + \omega \mathcal{L}_\vartheta(\mathcal{R}_{j_0}). \quad (\text{C.11})$$

Appendix C.1. Preparation: Energy Bound of Projection Operator

Each Bernoulli sample contributes a transformed ambient basis tensor to the golfing recursion. Bernstein inequalities therefore require a uniform bound on the energy of one such tensor after tangent-space projection. The next lemma provides this bound and is where the core incoherence μ and row incoherence μ_D enter the concentration estimates.

Lemma 3 (Projection norm of transformed ambient basis tensors). *Let $\mathcal{E}_{ijk}^{\text{orig}}$ be the (i, j, k) -th standard basis tensor of the original tensor space and define*

$$\mathcal{E}_{ijk}^{D_0} := \mathcal{A}_{D_0}^T(\mathcal{E}_{ijk}^{\text{orig}}) = \mathcal{E}_{ijk}^{\text{orig}} \times_3 D_0^T. \quad (\text{C.12})$$

Then

$$\|\mathcal{P}_\mathcal{T}(\mathcal{E}_{ijk}^{D_0})\|_F^2 \leq \frac{4\mu\mu_D R r_3}{n_{(2)}n_3}. \quad (\text{C.13})$$

Furthermore, for either sign $s \in \{-1, 1\}$,

$$\|\mathcal{P}_\mathcal{T} \mathcal{L}_\vartheta^s(\mathcal{E}_{ijk}^{D_0})\|_F^2 \leq \frac{4\kappa_\vartheta^2 \mu \mu_D R r_3}{n_{(2)}n_3} \max\{\vartheta_{\max}^2, \vartheta_{\min}^{-2}\}. \quad (\text{C.14})$$

Proof. For the skinny reduced-core t-SVD $\mathcal{B}_0 = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$, the tangent-space projection is

$$\mathcal{P}_\mathcal{T}(\mathcal{X}) = \mathcal{U} * \mathcal{U}^T * \mathcal{X} + \mathcal{X} * \mathcal{V} * \mathcal{V}^T - \mathcal{U} * \mathcal{U}^T * \mathcal{X} * \mathcal{V} * \mathcal{V}^T. \quad (\text{C.15})$$

Since the last term is the intersection of the first two orthogonal components,

$$\|\mathcal{P}_\mathcal{T}(\mathcal{X})\|_F^2 \leq 2\|\mathcal{U}^T * \mathcal{X}\|_F^2 + 2\|\mathcal{X} * \mathcal{V}\|_F^2. \quad (\text{C.16})$$

The transformed basis tensor has only one nonzero spatial position and its third-mode tube equals $D_0(k, :)^T$. Hence

$$\|\mathcal{E}_{ijk}^{D_0}\|_F^2 = \|D_0(k, :)\|_2^2 \leq \frac{\mu_D r_3}{n_3}. \quad (\text{C.17})$$

Using the two core-incoherence inequalities in Definition 6, Parseval's identity, and (C.17), we obtain

$$\|\mathcal{U}^T * \mathcal{E}_{ijk}^{D_0}\|_F^2 \leq \frac{\mu\mu_D R r_3}{n_1 n_3}, \quad (\text{C.18})$$

$$\|\mathcal{E}_{ijk}^{D_0} * \mathcal{V}\|_F^2 \leq \frac{\mu\mu_D R r_3}{n_2 n_3}. \quad (\text{C.19})$$

Substituting these estimates into (C.16) and using $n_{(2)} = \min\{n_1, n_2\}$ proves (C.13). The weighted estimate follows from (C.2)–(C.3). $\square$

The projection-energy bound controls individual random increments. We next identify their mean and show that their fluctuations contract the tangent residual at every golfing step.

Appendix C.2. Expectation and Variance Analysis of Iterative Update

For a Bernoulli set $\Lambda \sim \text{Ber}(p)$, define the weighted sampling operator on the reduced core by

$$\mathcal{M}_{\Lambda,p} := p^{-1} \mathcal{L}_\vartheta^{-1} \mathcal{A}_{D_0}^T \mathcal{P}_\Lambda^{\text{orig}} \mathcal{A}_{D_0} \mathcal{L}_\vartheta. \quad (\text{C.20})$$

The normalization in $\mathcal{M}_{\Lambda,p}$ makes sampling unbiased after lifting to the ambient space and returning to the reduced core. Hence the golfing residual is centered, which is the starting point for contraction.

Lemma 4 (Unbiasedness of the weighted sampling operator). *For every fixed reduced-core tensor $\mathcal{X}$,*

$$\mathbb{E}[\mathcal{M}_{\Lambda,p}(\mathcal{X})] = \mathcal{X}. \quad (\text{C.21})$$

Consequently, for the recursion in (C.5)–(C.6),

$$\mathbb{E}[\mathcal{R}_j \mid \mathcal{R}_{j-1}] = 0. \quad (\text{C.22})$$

Proof. Since $\mathbb{E}[\mathcal{P}_\Lambda^{\text{orig}}(\mathcal{Z})] = p\mathcal{Z}$,

$$\mathbb{E}[\mathcal{M}_{\Lambda,p}(\mathcal{X})] = \mathcal{L}_\vartheta^{-1} \mathcal{A}_{D_0}^T \mathcal{A}_{D_0} \mathcal{L}_\vartheta(\mathcal{X}) \quad (\text{C.23})$$

$$= \mathcal{L}_\vartheta^{-1} \mathcal{L}_\vartheta(\mathcal{X}) = \mathcal{X}, \quad (\text{C.24})$$

where $\mathcal{A}_{D_0}^T \mathcal{A}_{D_0} = \mathcal{I}$ was used. Since $\mathcal{R}_{j-1} \in \mathcal{T}$, the residual recursion is

$$\mathcal{R}_j = \mathcal{P}_\mathcal{T}(\mathcal{I} - \mathcal{M}_{\Gamma_j,q})\mathcal{P}_\mathcal{T}(\mathcal{R}_{j-1}),$$

and the conditional expectation is zero. $\square$

Unbiasedness does not by itself control a realized sampling operator. The next lemma upgrades it to high-probability operator and entrywise contraction. The operator estimate drives Frobenius contraction, while the entrywise estimate controls the infinity-norm budget of $\mathcal{F}^L$.

Lemma 5 (Weighted sampling contraction). *Let $0 < \tau < 1$. If*

$$p \geq C\tau^{-2}\kappa_{\vartheta}^2 \frac{\mu\mu_D Rr_3 \log(n_{(1)}n_3)}{n_{(2)}n_3}, \quad (\text{C.25})$$

then, with probability at least $1 - 2(n_{(1)}n_3)^{-11}$,

$$\|\mathcal{P}_{\mathcal{T}}(\mathcal{M}_{\Lambda,p} - \mathcal{I})\mathcal{P}_{\mathcal{T}}\|_{\text{op}} \leq \tau. \quad (\text{C.26})$$

For every fixed $\mathcal{X} \in \mathcal{T}$, the same event also yields

$$\|\mathcal{A}_{D_0}\mathcal{L}_{\vartheta}\mathcal{P}_{\mathcal{T}}(\mathcal{M}_{\Lambda,p} - \mathcal{I})(\mathcal{X})\|_{\infty} \leq \tau \|\mathcal{A}_{D_0}\mathcal{L}_{\vartheta}(\mathcal{X})\|_{\infty}. \quad (\text{C.27})$$

Proof. Write $a = (i, j, k)$ and let δ_a be the Bernoulli indicator of $a \in \Lambda$. Set

$$\sqcap_a := \mathcal{P}_{\mathcal{T}}\mathcal{L}_{\vartheta}^{-1}\mathcal{A}_{D_0}^T(\mathcal{E}_a^{\text{orig}}), \quad \sqsubseteq_a := \mathcal{P}_{\mathcal{T}}\mathcal{L}_{\vartheta}\mathcal{A}_{D_0}^T(\mathcal{E}_a^{\text{orig}}).$$

For the rank-one operator $(\sqcap \otimes \sqsubseteq)(\mathcal{X}) := \langle \sqsubseteq, \mathcal{X} \rangle \sqcap$, we have

$$\mathcal{P}_{\mathcal{T}}(\mathcal{M}_{\Lambda,p} - \mathcal{I})\mathcal{P}_{\mathcal{T}} = \sum_a \frac{\delta_a - p}{p} (\sqcap_a \otimes \sqsubseteq_a). \quad (\text{C.28})$$

The summands are independent and centered. By Lemma 3 and the weight-balance condition,

$$\|\sqcap_a\|_F \|\sqsubseteq_a\|_F \leq C\kappa_{\vartheta}^2 \frac{\mu\mu_D Rr_3}{n_{(2)}n_3} =: \nu. \quad (\text{C.29})$$

After applying the standard self-adjoint dilation, the operator norm of each dilated summand is at most ν/p , while the variance proxy is at most $C\nu/p$. Matrix Bernstein's inequality (Lemma 6) therefore gives

$$\Pr\left(\|\mathcal{P}_{\mathcal{T}}(\mathcal{M}_{\Lambda,p} - \mathcal{I})\mathcal{P}_{\mathcal{T}}\|_{\text{op}} > \tau\right) \leq 2(n_{(1)}n_3)^{-11} \quad (\text{C.30})$$

under (C.25).

For (C.27), fix an ambient coordinate a_0 and take the inner product of the sampled residual with $\mathcal{E}_{a_0}^{\text{orig}}$. This produces a sum of independent centered scalar random variables. Their maximal magnitude and variance are controlled by (C.13) and the core incoherence inequalities. Scalar Bernstein's inequality followed by a union bound over the $n_1n_2n_3$ ambient coordinates yields (C.27), after increasing the constant C in (C.25). $\square$

Appendix C.3. Finalizing the Proof of Golfing Scheme

Lemmas 3–5 now provide centered increments, geometric contraction, and uniform norm control. We combine them to finish the low-rank certificate construction.

Proof of Lemma 2. Choose $\tau = 1/2$ in Lemma 5. On the intersection of the j_0 concentration events,

$$\|\mathcal{R}_j\|_F \leq \frac{1}{2} \|\mathcal{R}_{j-1}\|_F, \quad \|\mathcal{A}_{D_0} \mathcal{L}_\vartheta(\mathcal{R}_j)\|_\infty \leq \frac{1}{2} \|\mathcal{A}_{D_0} \mathcal{L}_\vartheta(\mathcal{R}_{j-1})\|_\infty. \quad (\text{C.31})$$

Iterating the first inequality proves (C.10).

Since $\mathcal{W}^L = \mathcal{P}_{\mathcal{T}^\perp} \mathcal{Y}_{j_0}$, we immediately have $\mathcal{P}_{\mathcal{T}} \mathcal{W}^L = 0$. Moreover,

$$\mathcal{W}^L = \sum_{j=1}^{j_0} q^{-1} \mathcal{P}_{\mathcal{T}^\perp} \mathcal{L}_\vartheta^{-1} \mathcal{A}_{D_0}^T \mathcal{P}_{\Gamma_j}^{\text{orig}} \mathcal{A}_{D_0} \mathcal{L}_\vartheta(\mathcal{R}_{j-1}). \quad (\text{C.32})$$

Applying matrix Bernstein to each increment, using Lemma 3 and (C.31), gives

$$\|\mathcal{W}^L\| \leq C \left(\frac{\mu \mu_D R r_3 \log(n_{(1)} n_3)}{q n_{(2)} n_3} \right)^{1/2} \sum_{j=1}^{j_0} 2^{-(j-1)}. \quad (\text{C.33})$$

Under (C.8), the right-hand side is at most $1/8$ after increasing C_0 .

Because every Γ_j is contained in Γ^c , $\mathcal{P}_{\Gamma}^{\text{orig}} \mathcal{F}^L = 0$. Furthermore,

$$\|\mathcal{F}^L\|_\infty \leq \frac{\omega}{\lambda q} \sum_{j=1}^{j_0} \|\mathcal{A}_{D_0} \mathcal{L}_\vartheta(\mathcal{R}_{j-1})\|_\infty \quad (\text{C.34})$$

$$\leq \frac{2\omega}{\lambda q} \|\mathcal{A}_{D_0} \mathcal{L}_\vartheta(\mathcal{S}_0)\|_\infty. \quad (\text{C.35})$$

The modified incoherence condition and transform row incoherence imply

$$\|\mathcal{A}_{D_0} \mathcal{L}_\vartheta(\mathcal{S}_0)\|_\infty \leq C \vartheta_{\max} \sqrt{\frac{\mu \mu_D R r_3}{n_1 n_2 n_3}}. \quad (\text{C.36})$$

Using $\lambda = n_{(1)}^{-1/2}$, the rank condition in Theorem 1, and (C.8), the right-hand side of (C.35) is at most $1/4$ for a sufficiently small universal c_0 .

Finally, from (C.5) and (C.7),

$$\omega \mathcal{L}_\vartheta(\mathcal{Y}_{j_0}) = \lambda \mathcal{A}_{D_0}^T(\mathcal{F}^L). \quad (\text{C.37})$$

Since $\mathcal{S}_0 + \mathcal{W}^L = \mathcal{Y}_{j_0} + \mathcal{R}_{j_0}$, (C.11) follows. A union bound over $j_0 = O(\log n_{(1)})$ events gives the stated probability. $\square$

The golfing construction now has the correct tangent and support structure and has reduced the unresolved residual to $\mathcal{R}_{j_0}$. It does not yet incorporate $\text{sgn}(\mathcal{E}_0)$ into the certificate identity; Appendix D supplies that sparse-sign correction.

Corollary 2 (Low-rank certificate construction). *Under the conditions of Lemma 2, the tensors $(\mathcal{W}^L, \mathcal{F}^L, \mathcal{R}_{j_0})$ satisfy (C.9)–(C.11).*

For completeness, we record the general matrix and tensor identities used in the preceding concentration and mode-product calculations.

Lemma 6 (Matrix Bernstein inequality [3]). *Let $X_1, \dots, X_N \in \mathbb{R}^{d_1 \times d_2}$ be independent, centered random matrices satisfying $\|X_i\| \leq L$. Define*

$$\sigma^2 := \max \left\{ \left\| \sum_{i=1}^N \mathbb{E}(X_i X_i^T) \right\|, \left\| \sum_{i=1}^N \mathbb{E}(X_i^T X_i) \right\| \right\}.$$

Then, for every $t > 0$,

$$\Pr \left(\left\| \sum_{i=1}^N X_i \right\| \geq t \right) \leq (d_1 + d_2) \exp \left(-\frac{t^2/2}{\sigma^2 + Lt/3} \right). \quad (\text{C.38})$$

Lemma 7 (Frobenius norm invariance [4]). *For every $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times r_3}$,*

$$\|\mathcal{X} \times_3 D_0\|_F = \|\mathcal{X}\|_F. \quad (\text{C.39})$$

Proof. Using $D_0^T D_0 = I_{r_3}$,

$$\|\mathcal{X} \times_3 D_0\|_F^2 = \sum_{i,j,k} \left(\sum_{s=1}^{r_3} x_{ijs} D_0(k, s) \right)^2 \quad (\text{C.40})$$

$$= \sum_{i,j,s,t} x_{ijs} x_{ijt} [D_0^T D_0]_{st} = \|\mathcal{X}\|_F^2. \quad (\text{C.41})$$

$\square$

Lemma 8 (Matricization of a mode-3 product). *For $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and $D \in \mathbb{R}^{n_3 \times r_3}$,*

$$\text{unfold}_3(\mathcal{X} \times_3 D^T) = \text{unfold}_3(\mathcal{X})D. \quad (\text{C.42})$$

Proof. For every i, j , and $s \in [r_3]$, both sides have entry

$$\sum_{k=1}^{n_3} x_{ijk} D(k, s)$$

at row $i + (j - 1)n_1$ and column s . $\square$

Appendix D. Neumann Series Construction for $\mathcal{W}^S$

Appendix C produced a low-rank certificate with a small terminal residual, but the dual identity still lacks the sparse-sign term. This appendix constructs $(\mathcal{W}^S, \mathcal{F}^S)$ to insert $\text{sgn}(\mathcal{E}_0)$ and cancel that residual. The argument uses the least-squares/Neumann-series idea from robust low-rank recovery [11, 2], while keeping the ambient support projection and induced weighted core operator separate. The proof has three stages: invertibility of the complement tangent operator, an exact correction identity, and spectral/infinity norm control of the random correction.

Appendix D.1. Problem Setting and Theoretical Motivation

Let

$$p := 1 - \rho_s, \quad \mathcal{M}_c := p^{-1} \mathcal{L}_\vartheta^{-1} \mathcal{A}_{D_0}^T \mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{A}_{D_0} \mathcal{L}_\vartheta,$$

and define the tangent-space operator

$$\mathcal{B}_\Gamma := \mathcal{P}_\mathcal{T} \mathcal{M}_c \mathcal{P}_\mathcal{T} : \mathcal{T} \rightarrow \mathcal{T}. \quad (\text{D.1})$$

Let $\mathcal{R}_L := \mathcal{R}_{j_0}$ be the terminal golfing residual and set

$$\mathcal{Z}_S := \mathcal{R}_L - \frac{\lambda}{\omega} \mathcal{L}_\vartheta^{-1} \mathcal{P}_\mathcal{T} \mathcal{A}_{D_0}^T \text{sgn}(\mathcal{E}_0) \in \mathcal{T}. \quad (\text{D.2})$$

The target $\mathcal{Z}_S$ collects the two terms still missing after golfing: the terminal residual $\mathcal{R}_L$ and the tangent component of the sparse-sign tensor. Solving $\mathcal{B}_\Gamma \mathcal{X}_S = \mathcal{Z}_S$ therefore produces the correction needed to close the dual identity.

Definition 7 (Neumann-series sparse correction). *On the event $\|\mathcal{B}_\Gamma - \mathcal{P}_\mathcal{T}\|_{\text{op}} < 1$, define*

$$\mathcal{X}_S := \mathcal{B}_\Gamma^{-1} \mathcal{Z}_S = \sum_{k=0}^{\infty} (\mathcal{P}_\mathcal{T} - \mathcal{B}_\Gamma)^k \mathcal{Z}_S, \quad (\text{D.3})$$

$$\mathcal{Y}_S := \mathcal{M}_c(\mathcal{X}_S), \quad (\text{D.4})$$

$$\mathcal{W}^S := \mathcal{P}_{\mathcal{T}^\perp} \mathcal{Y}_S + \frac{\lambda}{\omega} \mathcal{L}_\vartheta^{-1} \mathcal{P}_{\mathcal{T}^\perp} \mathcal{A}_{D_0}^T \text{sgn}(\mathcal{E}_0), \quad (\text{D.5})$$

$$\mathcal{F}^S := \frac{\omega}{\lambda p} \mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{A}_{D_0} \mathcal{L}_\vartheta(\mathcal{X}_S). \quad (\text{D.6})$$

Appendix D.2. Convergence and Operator Norm Bound

The Neumann series is meaningful only if $\mathcal{B}_\Gamma$ is invertible on $\mathcal{T}$. The next lemma provides this inverse and its norm bound. The same concentration mechanism, without slice weighting, also yields the restricted-identifiability condition required by Lemma 1.

Lemma 9 (Invertibility of the complement tangent operator). *Under Conditions 2–3 of Theorem 1, for a sufficiently small universal constant c_0 , with probability at least $1 - 2(n_{(1)}n_3)^{-10}$,*

$$\|\mathcal{B}_\Gamma - \mathcal{P}_\mathcal{T}\|_{\text{op}} \leq \frac{1}{2}. \quad (\text{D.7})$$

Consequently, $\mathcal{B}_\Gamma$ is invertible on $\mathcal{T}$ and

$$\|\mathcal{B}_\Gamma^{-1}\|_{\text{op}} \leq 2. \quad (\text{D.8})$$

Proof. Since $\rho_s \leq c_0/(\mu_D r_3)$ and $\mu_D r_3 \geq 1$, decreasing c_0 if necessary gives $p = 1 - \rho_s \geq 3/4$. Apply Lemma 5 with $\Lambda = \Gamma^c$, sampling probability p , and $\tau = 1/2$. The required lower bound on p follows from the rank condition because

$$\frac{\mu \mu_D R r_3 \log(n_{(1)} n_3)}{n_{(2)} n_3} \leq C c_0 \frac{\mu_D r_3}{n_3 \log n_{(1)}} \leq C c_0,$$

where $\mu_D r_3 / n_3 \leq 1$ follows from $D_0^T D_0 = I_{r_3}$. This proves (D.7). The Neumann-series formula and (D.8) follow from

$$\mathcal{B}_\Gamma^{-1} = \sum_{k=0}^{\infty} (\mathcal{P}_\mathcal{T} - \mathcal{B}_\Gamma)^k \quad \text{on } \mathcal{T}.$$

□

Restricted identifiability. For completeness, define the corresponding unweighted complement operator

$$\widetilde{\mathcal{M}}_c := p^{-1} \mathcal{A}_{D_0}^T \mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{A}_{D_0}, \quad \widetilde{\mathcal{B}}_\Gamma := \mathcal{P}_\mathcal{T} \widetilde{\mathcal{M}}_c \mathcal{P}_\mathcal{T} : \mathcal{T} \rightarrow \mathcal{T}.$$

Applying the same concentration argument as in Lemma 5 with $\mathcal{L}_\vartheta = I$ gives, after enlarging the universal constant if necessary,

$$\left\| \widetilde{\mathcal{B}}_\Gamma - \mathcal{P}_\mathcal{T} \right\|_{\text{op}} \leq \frac{1}{2}$$

with probability at least $1 - 2(n_{(1)} n_3)^{-10}$. Consequently, $\widetilde{\mathcal{B}}_\Gamma$ is invertible on $\mathcal{T}$.

Now let $\mathcal{H} \in \mathcal{T}$ satisfy

$$\mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{A}_{D_0}(\mathcal{H}) = 0.$$

Then

$$\widetilde{\mathcal{B}}_\Gamma(\mathcal{H}) = p^{-1} \mathcal{P}_\mathcal{T} \mathcal{A}_{D_0}^T \mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{A}_{D_0}(\mathcal{H}) = 0.$$

The invertibility of $\widetilde{\mathcal{B}}_\Gamma$ on $\mathcal{T}$ therefore implies $\mathcal{H} = 0$. Hence the restricted-identifiability condition in Lemma 1 holds on the same high-probability event.

Restricted identifiability prevents a nonzero tangent perturbation from being hidden entirely on the corruption support. We now verify the deterministic algebraic properties of the correction, especially the identity that cancels the terminal golfing residual.

Appendix D.3. Norm Estimation

Lemma 10 (Algebraic properties of the Neumann correction). *On the event in Lemma 9,*

$$\|\mathcal{X}_S\|_F \leq 2\|\mathcal{Z}_S\|_F, \quad \mathcal{P}_\mathcal{T} \mathcal{W}^S = 0, \quad \mathcal{P}_\Gamma^{\text{orig}} \mathcal{F}^S = 0. \quad (\text{D.9})$$

Moreover,

$$\omega \mathcal{L}_\vartheta(\mathcal{R}_L + \mathcal{W}^S) = \lambda \mathcal{A}_{D_0}^T (\text{sgn}(\mathcal{E}_0) + \mathcal{F}^S). \quad (\text{D.10})$$

Proof. The inverse bound gives $\|\mathcal{X}_S\|_F \leq 2\|\mathcal{Z}_S\|_F$. By definition, $\mathcal{W}^S \in \mathcal{T}^\perp$ and $\mathcal{F}^S$ is supported on Γ^c . Since $\mathcal{B}_\Gamma \mathcal{X}_S = \mathcal{Z}_S$,

$$\mathcal{P}_\mathcal{T} \mathcal{Y}_S = \mathcal{P}_\mathcal{T} \mathcal{M}_c \mathcal{P}_\mathcal{T}(\mathcal{X}_S) = \mathcal{Z}_S.$$

Thus $\mathcal{Y}_S = \mathcal{Z}_S + \mathcal{P}_{\mathcal{T}^\perp} \mathcal{Y}_S$. Using (D.2) and (D.5),

$$\mathcal{R}_L + \mathcal{W}^S = \mathcal{Y}_S + \frac{\lambda}{\omega} \mathcal{L}_\vartheta^{-1} \mathcal{A}_{D_0}^T \text{sgn}(\mathcal{E}_0). \quad (\text{D.11})$$

Furthermore, by the definitions of $\mathcal{M}_c$ and $\mathcal{F}^S$,

$$\omega \mathcal{L}_\vartheta(\mathcal{Y}_S) = \frac{\omega}{p} \mathcal{A}_{D_0}^T \mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{A}_{D_0} \mathcal{L}_\vartheta(\mathcal{X}_S) = \lambda \mathcal{A}_{D_0}^T(\mathcal{F}^S). \quad (\text{D.12})$$

Applying $\omega \mathcal{L}_\vartheta$ to (D.11) proves (D.10). $\square$

Appendix D.4. Key Estimation of the Sparse-Sign Correction

The exact identity is not enough: Lemma 1 also requires $\|\mathcal{W}^S\| \leq 1/8$ and $\|\mathcal{F}^S\|_\infty \leq 1/4$. The next lemma proves these budgets by conditioning on the support and applying matrix and scalar concentration to the independent Rademacher signs.

Lemma 11 (Norm bounds for the sparse-sign correction). *Under Conditions 1–3 of Theorem 1, conditional on Γ and on the complement-operator event in Lemma 9, the independent Rademacher signs of $\mathcal{E}_0$ satisfy, with conditional probability at least $1 - C(n_1 n_3)^{-10}$,*

$$\|\mathcal{W}^S\| \leq \frac{1}{8}, \quad \|\mathcal{F}^S\|_\infty \leq \frac{1}{4}. \quad (\text{D.13})$$

Proof. Condition on the realization of Γ and on the event (D.7). Let

$$\mathcal{S}_\Gamma := \text{sgn}(\mathcal{E}_0) = \sum_{a \in \Gamma} \epsilon_a \mathcal{E}_a^{\text{orig}},$$

where the ϵ_a are independent symmetric Rademacher variables. Since $\mathcal{B}_\Gamma^{-1}$ and $\mathcal{M}_c$ depend only on Γ , they are deterministic under this conditioning. Decompose

$$\mathcal{X}_S = \mathcal{X}_R - \frac{\lambda}{\omega} \sum_{a \in \Gamma} \epsilon_a \mathcal{B}_\Gamma^{-1} \mathcal{L}_\vartheta^{-1} \mathcal{P}_\mathcal{T} \mathcal{A}_{D_0}^T(\mathcal{E}_a^{\text{orig}}), \quad (\text{D.14})$$

$$\mathcal{X}_R := \mathcal{B}_\Gamma^{-1} \mathcal{R}_L. \quad (\text{D.15})$$

Substitution into (D.5) gives

$$\mathcal{W}^S = \mathcal{W}_R + \sum_{a \in \Gamma} \epsilon_a \mathcal{G}_a, \quad (\text{D.16})$$

where

$$\mathcal{W}_R := \mathcal{P}_{\mathcal{T}^\perp} \mathcal{M}_c(\mathcal{X}_R), \quad (\text{D.17})$$

$$\mathcal{G}_a := \frac{\lambda}{\omega} \left[\mathcal{L}_\vartheta^{-1} \mathcal{P}_{\mathcal{T}^\perp} \mathcal{A}_{D_0}^T(\mathcal{E}_a^{\text{orig}}) - \mathcal{P}_{\mathcal{T}^\perp} \mathcal{M}_c \mathcal{B}_\Gamma^{-1} \mathcal{L}_\vartheta^{-1} \mathcal{P}_\mathcal{T} \mathcal{A}_{D_0}^T(\mathcal{E}_a^{\text{orig}}) \right]. \quad (\text{D.18})$$

Similarly,

$$\mathcal{F}^S = \mathcal{F}_R + \sum_{a \in \Gamma} \epsilon_a \mathcal{H}_a, \quad (\text{D.19})$$

where

$$\mathcal{F}_R := \frac{\omega}{\lambda p} \mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{A}_{D_0} \mathcal{L}_\vartheta(\mathcal{X}_R), \quad (\text{D.20})$$

$$\mathcal{H}_a := -\frac{1}{p} \mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{A}_{D_0} \mathcal{L}_\vartheta \mathcal{B}_\Gamma^{-1} \mathcal{L}_\vartheta^{-1} \mathcal{P}_\mathcal{T} \mathcal{A}_{D_0}^T(\mathcal{E}_a^{\text{orig}}). \quad (\text{D.21})$$

Step 1: Deterministic terminal-residual terms. By (D.8) and (C.10),

$$\|\mathcal{X}_R\|_F \leq 2\|\mathcal{R}_L\|_F \leq 2n_{(1)}^{-9} \|\mathcal{S}_0\|_F.$$

The operator bounds in Lemmas 3 and 9 therefore imply

$$\|\mathcal{W}_R\| + \|\mathcal{F}_R\|_\infty \leq Cn_{(1)}^{-8}. \quad (\text{D.22})$$

Step 2: Spectral norm of the Rademacher tensor series. Let $\widehat{\mathcal{G}}_a^{(\ell)}$ denote the ℓ -th Fourier slice of $\mathcal{G}_a$. Lemma 3, $\|\mathcal{B}_\Gamma^{-1}\|_{\text{op}} \leq 2$, and $\|\mathcal{M}_c\|_{\text{op}} \leq C$ on the complement-operator event give

$$\max_{a, \ell} \|\widehat{\mathcal{G}}_a^{(\ell)}\| \leq C \frac{\lambda \kappa_\vartheta}{\omega} \sqrt{\frac{\mu_D r_3}{n_3}}. \quad (\text{D.23})$$

Using the Parseval-frame identity

$$\sum_{i,j,k} \mathcal{A}_{D_0}^T(\mathcal{E}_{ijk}^{\text{orig}}) \otimes \mathcal{A}_{D_0}^T(\mathcal{E}_{ijk}^{\text{orig}}) = \mathcal{I}$$

and the tangent-space energy estimate (C.13), the row and column variance parameters of each Fourier-slice series satisfy

$$\max \left\{ \left\| \sum_{a \in \Gamma} \widehat{\mathcal{G}}_a^{(\ell)} \widehat{\mathcal{G}}_a^{(\ell)T} \right\|, \left\| \sum_{a \in \Gamma} \widehat{\mathcal{G}}_a^{(\ell)T} \widehat{\mathcal{G}}_a^{(\ell)} \right\| \right\} \leq C \frac{\lambda^2 \kappa_\vartheta^2}{\omega^2} \mu_D \rho_s n_{(1)} r_3. \quad (\text{D.24})$$

Matrix Bernstein's inequality, followed by a union bound over $\ell = 1, \dots, r_3$, now yields

$$\left\| \sum_{a \in \Gamma} \epsilon_a \mathcal{G}_a \right\| \leq C \frac{\lambda \kappa_{\vartheta}}{\omega} \left[\sqrt{\mu_D \rho_s n_{(1)} r_3} + \sqrt{\frac{\mu_D r_3}{n_3}} \log(n_{(1)} n_3) \right] \quad (\text{D.25})$$

with conditional probability at least $1 - C(n_{(1)} n_3)^{-11}$. Since $\lambda = n_{(1)}^{-1/2}$ and $\mu_D \rho_s r_3 \leq c_0$, the first term is bounded by $C \kappa_{\vartheta} \sqrt{c_0}/\omega$. The second term is of lower order under the same dimension regime used in the tensor concentration bounds. Thus, after decreasing c_0 if necessary and combining with (D.22),

$$\|\mathcal{W}^S\| \leq \frac{1}{8}. \quad (\text{D.26})$$

Step 3: Infinity norm of the ambient Rademacher series. Fix an ambient coordinate b . From (D.21), $[\sum_{a \in \Gamma} \epsilon_a \mathcal{H}_a]_b$ is a scalar Rademacher series. Lemma 3 and the inverse bound imply

$$\sum_{a \in \Gamma} |[\mathcal{H}_a]_b|^2 \leq C \kappa_{\vartheta}^2 \frac{\mu \mu_D \rho_s R r_3}{n_{(2)}}, \quad (\text{D.27})$$

$$\max_{a,b} |[\mathcal{H}_a]_b| \leq C \kappa_{\vartheta} \frac{\mu \mu_D R r_3}{n_{(2)} n_3}. \quad (\text{D.28})$$

Scalar Bernstein's inequality and a union bound over all $n_1 n_2 n_3$ ambient coordinates give

$$\begin{aligned} \left\| \sum_{a \in \Gamma} \epsilon_a \mathcal{H}_a \right\|_{\infty} &\leq C \kappa_{\vartheta} \sqrt{\frac{\mu \mu_D \rho_s R r_3 \log(n_{(1)} n_3)}{n_{(2)}}} \\ &\quad + C \kappa_{\vartheta} \frac{\mu \mu_D R r_3 \log(n_{(1)} n_3)}{n_{(2)} n_3} \end{aligned} \quad (\text{D.29})$$

with conditional probability at least $1 - C(n_{(1)} n_3)^{-11}$. By Conditions 2 and 3, the first term is at most $C c_0 / \sqrt{\log n_{(1)}}$, and the second is at most $C c_0 / \log n_{(1)}$, up to an immaterial adjustment of the universal logarithmic constant. Hence, for sufficiently small c_0 , combining (D.29) with (D.22) gives

$$\|\mathcal{F}^S\|_{\infty} \leq \frac{1}{4}. \quad (\text{D.30})$$

The union of the two conditional concentration events proves (D.13). $\square$

The sparse-sign correction now satisfies both the exact cancellation identity and the remaining norm budgets. Appendices C and D have produced all pieces required by Appendix B. Appendix E combines them and closes the proof of Theorem 1.

Appendix E. Verification of Dual Proof Conditions

This appendix is the final stage of the recovery proof. The golfing component provides $(\mathcal{W}^L, \mathcal{F}^L)$ and a small residual, while the Neumann component provides $(\mathcal{W}^S, \mathcal{F}^S)$ and cancels that residual together with the sparse-sign term. The next lemma verifies that their sum satisfies every hypothesis of Lemma 1 on one high-probability event.

Lemma 12 (Verification of dual-certificate conditions). *Let*

$$\mathcal{W} := \mathcal{W}^L + \mathcal{W}^S, \quad \mathcal{F} := \mathcal{F}^L + \mathcal{F}^S, \quad \mathcal{D} := 0. \quad (\text{E.1})$$

Under the assumptions of Theorem 1, with probability at least $1 - c_1(n_{(1)}n_3)^{-c_2}$, the triple $(\mathcal{W}, \mathcal{F}, \mathcal{D})$ satisfies all conditions in Lemma 1.

Proof. Adding the exact identities (C.11) and (D.10) cancels the terminal residual and gives

$$\omega \mathcal{L}_\vartheta(\mathcal{S}_0 + \mathcal{W}) = \lambda \mathcal{A}_{D_0}^T (\text{sgn}(\mathcal{E}_0) + \mathcal{F}^L + \mathcal{F}^S) \quad (\text{E.2})$$

$$= \lambda \mathcal{A}_{D_0}^T (\text{sgn}(\mathcal{E}_0) + \mathcal{F} + \mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{D}), \quad (\text{E.3})$$

which is exactly the dual-feasibility identity (B.4), because $\mathcal{L}_\vartheta(\mathcal{X}) = \vartheta \otimes \mathcal{X}$ and $\mathcal{D} = 0$.

Both $\mathcal{W}^L$ and $\mathcal{W}^S$ lie in $\mathcal{T}^\perp$; hence

$$\mathcal{P}_{\mathcal{T}} \mathcal{W} = 0.$$

Lemmas 2 and 11 give

$$\|\mathcal{W}\| \leq \|\mathcal{W}^L\| + \|\mathcal{W}^S\| \leq \frac{1}{8} + \frac{1}{8} = \frac{1}{4}.$$

Similarly, both $\mathcal{F}^L$ and $\mathcal{F}^S$ are supported on Γ^c , so

$$\mathcal{P}_{\Gamma}^{\text{orig}} \mathcal{F} = 0, \quad \|\mathcal{F}\|_\infty \leq \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

Finally,

$$\|\mathcal{P}_{\Gamma^c}^{\text{orig}} \mathcal{D}\|_F = 0 \leq \frac{\sqrt{r_3}}{4}.$$

Thus all conditions in (B.5) hold. Moreover, the unweighted complement-operator estimate established in Lemma 9 verifies restricted identifiability in (B.3). A union bound over the golfing, complement-operator, and conditional Rademacher events yields the stated probability and completes the verification. $\square$

Completion of the proof of Theorem 1. On the event in Lemma 12, the combined tensors $(\mathcal{W}, \mathcal{F}, \mathcal{D})$ satisfy the dual-feasibility identity, the tangent-space and support conditions, the required norm bounds, and restricted identifiability. Lemma 1 therefore gives an objective gap greater than $-\delta$ for every nonzero feasible perturbation with $\|\mathcal{H}\|_F \leq \epsilon$, where

$$\delta = \epsilon \left(\frac{\lambda \sqrt{r_3}}{4} + \vartheta_{\max} \tilde{\omega}_{\max} r r_3 \right).$$

The union bound gives probability at least $1 - c_1(n_{(1)}n_3)^{-c_2}$. Hence $(\mathcal{B}_0, \mathcal{E}_0)$ is an (ϵ, δ) -asymptotically unique local solution of problem (A.12), which proves Theorem 1. This is the end of the recovery proof. $\square$

Appendix F. Algorithm Convergence Analysis

In this section, we establish the analytical convergence properties of the ADMM-type scheme used to solve the AC²-TRPCA model. Because the objective contains a nonconvex convex-concave regularizer and D is constrained to the Stiefel manifold, standard convex ADMM theory does not directly apply. We therefore use subproblem optimality, multiplier estimates, and limiting-subdifferential arguments.

Specifically, we first prove boundedness of the Lagrange multiplier sequence, then establish convergence of the primal feasibility residual, and finally give a conditional accumulation-point result: under explicit boundedness and asymptotic-regularity assumptions, every accumulation point satisfies generalized Karush–Kuhn–Tucker (KKT) conditions of the original nonconvex optimization problem. Throughout this appendix, the asymptotic convergence analysis is conducted under the uncapped penalty schedule

$$\mu^{(k+1)} = \rho \mu^{(k)}, \quad \rho > 1,$$

which corresponds to setting $\mu_{\max} = +\infty$ in Algorithm 1 of the main paper. The finite penalty cap used in the numerical implementation serves only as a stabilization safeguard.

Appendix F.1. Augmented Lagrangian and Iteration Rules

For completeness, we first restate the augmented Lagrangian function and the iteration rules of our algorithm. The augmented Lagrangian function for the AC²-TRPCA model is defined as:

$$\begin{aligned} \mathcal{L}(\mathcal{B}, D, \mathcal{E}, \Lambda, \mu) = & \text{Sep}(\mathcal{B}, \Omega, \vartheta) + \lambda \|\mathcal{E}\|_1 \\ & + \langle \Lambda, \mathcal{Y} - \mathcal{B} \times_3 D - \mathcal{E} \rangle + \frac{\mu}{2} \|\mathcal{Y} - \mathcal{B} \times_3 D - \mathcal{E}\|_F^2. \end{aligned} \quad (\text{F.1})$$

At the k -th iteration, the algorithm sequentially updates the variables as follows:

$$\mathcal{B}^{(k+1)} \in \arg \min_{\mathcal{B}} \text{Sep}(\mathcal{B}, \Omega, \vartheta) + \frac{\mu^{(k)}}{2} \left\| \mathcal{B} - \left(\mathcal{Y} - \mathcal{E}^{(k)} + \frac{\Lambda^{(k)}}{\mu^{(k)}} \right) \times_3 (D^{(k)})^T \right\|_F^2, \quad (\text{F.2})$$

$$D^{(k+1)} \in \arg \max_{D^T D = I_{r_3}} \text{tr} \left(D^T M^{(k)} \right), \quad (\text{F.3})$$

$$\mathcal{E}^{(k+1)} = \arg \min_{\mathcal{E}} \lambda \|\mathcal{E}\|_1 + \frac{\mu^{(k)}}{2} \left\| \mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} + \frac{\Lambda^{(k)}}{\mu^{(k)}} - \mathcal{E} \right\|_F^2, \quad (\text{F.4})$$

$$\Lambda^{(k+1)} = \Lambda^{(k)} + \mu^{(k)} \left(\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)} \right), \quad (\text{F.5})$$

where $M^{(k)} = [\text{unfold}_3(\mathcal{Y} - \mathcal{E}^{(k)} + \Lambda^{(k)}/\mu^{(k)})]^T \cdot \text{unfold}_3(\mathcal{B}^{(k+1)})$.

For a matrix $M \in \mathbb{R}^{n_3 \times r_3}$ with a thin SVD $M = U_M \Sigma_M V_M^T$, define the SVD-based polar-factor set by

$$\text{Polar}(M) := \{U_M V_M^T : M = U_M \Sigma_M V_M^T \text{ is a thin SVD}\}. \quad (\text{F.6})$$

If M has full column rank, this set is a singleton and $\text{Polar}(M) = M(M^T M)^{-1/2}$. If M is rank deficient, any element of the above set is an admissible global solution of the rectangular orthogonal Procrustes subproblem.

Appendix F.2. Preliminaries for Convergence

Before presenting the main theorems, we provide the basic definitions and lemmas from variational analysis needed for the proofs.

Definition 8 (Limiting Subdifferential [10]). *Let $G : \mathbb{R}^{n_1 \times n_2 \times n_3} \rightarrow \mathbb{R}$ be a proper lower semi-continuous function. The limiting subdifferential of G at $\mathcal{X}$ is defined as:*

$$\partial G(\mathcal{X}) = \left\{ \mathcal{Z} : \exists \mathcal{X}^{(k)} \rightarrow \mathcal{X}, G(\mathcal{X}^{(k)}) \rightarrow G(\mathcal{X}), \mathcal{Z}^{(k)} \in \hat{\partial} G(\mathcal{X}^{(k)}) \rightarrow \mathcal{Z} \right\}, \quad (\text{F.7})$$

where $\hat{\partial} G(\mathcal{X})$ is the Fréchet subdifferential.

Lemma 13 (Fermat's Rule [10]). *If $\mathcal{X}$ is a local minimum of G , then $0 \in \partial G(\mathcal{X})$.*

Lemma 14 (Sum Rule [10]). *If f is continuously differentiable and G is lower semi-continuous, then*

$$\partial(f + G)(\mathcal{X}) = \nabla f(\mathcal{X}) + \partial G(\mathcal{X}). \quad (\text{F.8})$$

Lemma 15 (Outer Semicontinuity [10]). *Let $\mathcal{X}^{(k)} \rightarrow \mathcal{X}$, $G(\mathcal{X}^{(k)}) \rightarrow G(\mathcal{X})$, $\mathcal{Z}^{(k)} \in \partial G(\mathcal{X}^{(k)})$, and $\mathcal{Z}^{(k)} \rightarrow \mathcal{Z}$. Then $\mathcal{Z} \in \partial G(\mathcal{X})$.*

Appendix F.3. Proofs of Convergence Theorems

First, we prove that the sequence of Lagrange multipliers generated by the algorithm is bounded.

Theorem 2 (Boundedness of Lagrange Multipliers). *The sequence of Lagrange multipliers $\{\Lambda^{(k)}\}_{k \geq 1}$ generated by the algorithm is bounded. Specifically, for any $k \geq 1$:*

$$\|\Lambda^{(k)}\|_{\infty} \leq \lambda, \quad \text{and} \quad \|\Lambda^{(k)}\|_F \leq \sqrt{n_1 n_2 n_3} \lambda. \quad (\text{F.9})$$

Proof. From the first-order optimality condition (Fermat's Rule, Lemma 13) for the $\mathcal{E}$ -subproblem in (F.4):

$$0 \in \lambda \partial \|\mathcal{E}^{(k+1)}\|_1 - \mu^{(k)} \left(\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)} + \frac{\Lambda^{(k)}}{\mu^{(k)}} \right). \quad (\text{F.10})$$

Rearranging yields:

$$\Lambda^{(k)} + \mu^{(k)} (\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)}) \in \lambda \partial \|\mathcal{E}^{(k+1)}\|_1. \quad (\text{F.11})$$

According to the multiplier update formula (F.5), the left-hand side is exactly $\Lambda^{(k+1)}$:

$$\Lambda^{(k+1)} \in \lambda \partial \|\mathcal{E}^{(k+1)}\|_1. \quad (\text{F.12})$$

By the properties of the subdifferential of the ℓ_1 -norm, for any $G \in \partial\|\mathcal{X}\|_1$, we have $\|G\|_\infty \leq 1$. Therefore:

$$\|\Lambda^{(k+1)}\|_\infty \leq \lambda. \quad (\text{F.13})$$

By the equivalence of norms in finite-dimensional spaces:

$$\|\Lambda^{(k+1)}\|_F \leq \sqrt{n_1 n_2 n_3} \|\Lambda^{(k+1)}\|_\infty \leq \sqrt{n_1 n_2 n_3} \lambda =: C_0. \quad (\text{F.14})$$

Hence, $\{\Lambda^{(k)}\}_{k \geq 1}$ is bounded. $\square$

Building upon the boundedness of the multipliers, we next establish that the primal feasibility residual sequence converges to zero.

Theorem 3 (Convergence of the Primal Residual). *Under the assumptions of Theorem 2, suppose that the penalty parameter follows the uncapped update*

$$\mu^{(k+1)} = \rho \mu^{(k)}, \quad \rho > 1, \quad \mu^{(0)} > 0.$$

Then the primal feasibility residual converges to zero:

$$\lim_{k \rightarrow \infty} \|\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)}\|_F = 0. \quad (\text{F.15})$$

Proof. Define

$$\mathcal{R}^{(k+1)} := \mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)}.$$

From the multiplier update

$$\Lambda^{(k+1)} = \Lambda^{(k)} + \mu^{(k)} \mathcal{R}^{(k+1)},$$

we obtain

$$\mathcal{R}^{(k+1)} = \frac{\Lambda^{(k+1)} - \Lambda^{(k)}}{\mu^{(k)}}.$$

By Theorem 2, there exists $C_0 > 0$ such that $\|\Lambda^{(k)}\|_F \leq C_0$ for all k . Hence,

$$\|\mathcal{R}^{(k+1)}\|_F \leq \frac{\|\Lambda^{(k+1)}\|_F + \|\Lambda^{(k)}\|_F}{\mu^{(k)}} \leq \frac{2C_0}{\mu^{(k)}}.$$

Since

$$\mu^{(k)} = \mu^{(0)} \rho^k \longrightarrow +\infty,$$

it follows that

$$\|\mathcal{R}^{(k+1)}\|_F \leq \frac{2C_0}{\mu^{(0)}} \rho^{-k} \longrightarrow 0.$$

$\square$

Remark 2. *The above asymptotic result is established for the uncapped theoretical schedule $\mu_{\max} = +\infty$. A finite $\mu_{\max}$ is used only as a numerical safeguard in the implementation and is assessed through the residual-based stopping criterion.*

Because the convex–concave regularizer is noncoercive in general, boundedness of the complete primal–dual sequence does not follow from the objective value alone. We therefore state boundedness and the scaled asymptotic-regularity conditions explicitly in the following accumulation-point result.

Theorem 4 (Generalized KKT Property of Accumulation Points). *Under the assumptions of Theorem 3, define*

$$Z^{(k)} := (\mathcal{B}^{(k)}, D^{(k)}, \mathcal{E}^{(k)}, \Lambda^{(k)}).$$

Suppose that $\{Z^{(k)}\}_{k \geq 0}$ is bounded and satisfies

$$\|\Delta \mathcal{B}^{(k)}\|_F + \|\Delta \Lambda^{(k)}\|_F + \mu^{(k)} (\|\Delta D^{(k)}\|_F + \|\Delta \mathcal{E}^{(k)}\|_F) \rightarrow 0, \quad (\text{F.16})$$

where $\Delta X^{(k)} := X^{(k+1)} - X^{(k)}$. Then $\{Z^{(k)}\}_{k \geq 0}$ admits at least one accumulation point, and every accumulation point $Z^ := (\mathcal{B}^*, D^*, \mathcal{E}^*, \Lambda^*)$ satisfies*

$$\left\{ \begin{array}{l} 0 \in \partial \text{Sep}(\mathcal{B}^*, \Omega, \vartheta) - \Lambda^* \times_3 (D^*)^T, \\ 0 \in \lambda \partial \|\mathcal{E}^*\|_1 - \Lambda^*, \\ 0 \in -\text{unfold}_3(\Lambda^*)^T \text{unfold}_3(\mathcal{B}^*) + N_{\text{St}}(D^*), \\ \mathcal{Y} = \mathcal{B}^* \times_3 D^* + \mathcal{E}^*, \\ (D^*)^T D^* = I_{r_3}. \end{array} \right. \quad (\text{F.17})$$

Here, $N_{\text{St}}(D^)$ denotes the normal cone to $\text{St}(n_3, r_3) := \{D \in \mathbb{R}^{n_3 \times r_3} : D^T D = I_{r_3}\}$ at D^* .*

Proof. (I) Subsequence convergence

By the boundedness assumption, the sequence $\{(\mathcal{B}^{(k)}, D^{(k)}, \mathcal{E}^{(k)}, \Lambda^{(k)})\}$ has a convergent subsequence, denoted by $\{k_j\}$, such that

$$(\mathcal{B}^{(k_j)}, D^{(k_j)}, \mathcal{E}^{(k_j)}, \Lambda^{(k_j)}) \rightarrow (\mathcal{B}^*, D^*, \mathcal{E}^*, \Lambda^*). \quad (\text{F.18})$$

Condition (F.16) also implies that the corresponding one-step-shifted primal–dual variables converge to the same limit. Theorem 2 supplies the required uniform bound for the multiplier sequence.

(II) Stationarity with respect to $\mathcal{B}$

Step 1: First-order optimality condition of the B-subproblem

The update rule for the $\mathcal{B}$ -subproblem (F.2) at iteration k implies, by Fermat's Rule (Lemma 13):

$$0 \in \partial \left(\text{Sep}(\mathcal{B}^{(k+1)}, \Omega, \vartheta) + \frac{\mu^{(k)}}{2} \|\mathcal{B}^{(k+1)} - \mathcal{N}^{(k)}\|_F^2 \right), \quad (\text{F.19})$$

where $\mathcal{N}^{(k)} = (\mathcal{Y} - \mathcal{E}^{(k)} + \frac{\Lambda^{(k)}}{\mu^{(k)}}) \times_3 (D^{(k)})^T$.

Step 2: Applying the subdifferential sum rule Since the quadratic term $\frac{\mu^{(k)}}{2} \|\mathcal{B} - \mathcal{N}^{(k)}\|_F^2$ is differentiable with respect to $\mathcal{B}$ with gradient $\mu^{(k)}(\mathcal{B}^{(k+1)} - \mathcal{N}^{(k)})$, applying the sum rule (Lemma 14):

$$0 \in \partial \text{Sep}(\mathcal{B}^{(k+1)}, \Omega, \vartheta) + \mu^{(k)} (\mathcal{B}^{(k+1)} - \mathcal{N}^{(k)}). \quad (\text{F.20})$$

Substituting the definition of $\mathcal{N}^{(k)}$:

$$0 \in \partial \text{Sep}(\mathcal{B}^{(k+1)}, \Omega, \vartheta) + \mu^{(k)} \mathcal{B}^{(k+1)} - \mu^{(k)} \left(\mathcal{Y} - \mathcal{E}^{(k)} + \frac{\Lambda^{(k)}}{\mu^{(k)}} \right) \times_3 (D^{(k)})^T. \quad (\text{F.21})$$

Step 3: Rearranging terms Rearranging to isolate the subgradient term:

$$\Lambda^{(k)} \times_3 (D^{(k)})^T \in \partial \text{Sep}(\mathcal{B}^{(k+1)}, \Omega, \vartheta) + \mu^{(k)} [\mathcal{B}^{(k+1)} - (\mathcal{Y} - \mathcal{E}^{(k)}) \times_3 (D^{(k)})^T]. \quad (\text{F.22})$$

Using

$$(\mathcal{B}^{(k+1)} \times_3 D^{(k)}) \times_3 (D^{(k)})^T = \mathcal{B}^{(k+1)},$$

which follows from $(D^{(k)})^T D^{(k)} = I_{r_3}$, we rewrite the bracketed term:

$$\mathcal{B}^{(k+1)} - (\mathcal{Y} - \mathcal{E}^{(k)}) \times_3 (D^{(k)})^T = [\mathcal{B}^{(k+1)} \times_3 D^{(k)} + \mathcal{E}^{(k)} - \mathcal{Y}] \times_3 (D^{(k)})^T. \quad (\text{F.23})$$

Thus, Eq. (F.22) becomes:

$$\begin{aligned} \Lambda^{(k)} \times_3 (D^{(k)})^T &\in \partial \text{Sep}(\mathcal{B}^{(k+1)}, \Omega, \vartheta) \\ &+ \mu^{(k)} [\mathcal{B}^{(k+1)} \times_3 D^{(k)} + \mathcal{E}^{(k)} - \mathcal{Y}] \times_3 (D^{(k)})^T. \end{aligned} \quad (\text{F.24})$$

Step 4: Introducing variable updates and dual residuals Define the difference in variables between successive iterations as:

$$\Delta D^{(k)} = D^{(k+1)} - D^{(k)}, \quad \Delta \mathcal{E}^{(k)} = \mathcal{E}^{(k+1)} - \mathcal{E}^{(k)}. \quad (\text{F.25})$$

Rewriting $\mathcal{B}^{(k+1)} \times_3 D^{(k)} + \mathcal{E}^{(k)} - \mathcal{Y}$ using the updated variables $(k+1)$:

$$\begin{aligned}
& \mathcal{B}^{(k+1)} \times_3 D^{(k)} + \mathcal{E}^{(k)} - \mathcal{Y} \\
&= [\mathcal{B}^{(k+1)} \times_3 D^{(k+1)} + \mathcal{E}^{(k+1)} - \mathcal{Y}] \\
&\quad + [\mathcal{B}^{(k+1)} \times_3 (D^{(k)} - D^{(k+1)}) + (\mathcal{E}^{(k)} - \mathcal{E}^{(k+1)})] \\
&= [\mathcal{B}^{(k+1)} \times_3 D^{(k+1)} + \mathcal{E}^{(k+1)} - \mathcal{Y}] - [\mathcal{B}^{(k+1)} \times_3 \Delta D^{(k)} + \Delta \mathcal{E}^{(k)}].
\end{aligned} \tag{F.26}$$

Step 5: Applying the Lagrangian multiplier update rule Based on the multiplier update rule (F.5):

$$\Lambda^{(k+1)} = \Lambda^{(k)} + \mu^{(k)} (\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)}). \tag{F.27}$$

This implies:

$$\mu^{(k)} [\mathcal{B}^{(k+1)} \times_3 D^{(k+1)} + \mathcal{E}^{(k+1)} - \mathcal{Y}] = \Lambda^{(k)} - \Lambda^{(k+1)}. \tag{F.28}$$

Substituting this back into Eq. (F.24) and letting $\xi^{(k)} \in \partial \text{Sep}(\mathcal{B}^{(k+1)}, \Omega, \vartheta)$:

$$\begin{aligned}
\Lambda^{(k)} \times_3 (D^{(k)})^T &= \xi^{(k)} + (\Lambda^{(k)} - \Lambda^{(k+1)}) \times_3 (D^{(k)})^T \\
&\quad - \mu^{(k)} [\mathcal{B}^{(k+1)} \times_3 \Delta D^{(k)} + \Delta \mathcal{E}^{(k)}] \times_3 (D^{(k)})^T.
\end{aligned} \tag{F.29}$$

Simplifying ($\Lambda^{(k)}$ terms cancel):

$$\Lambda^{(k+1)} \times_3 (D^{(k)})^T = \xi^{(k)} - \mu^{(k)} [\mathcal{B}^{(k+1)} \times_3 \Delta D^{(k)} + \Delta \mathcal{E}^{(k)}] \times_3 (D^{(k)})^T. \tag{F.30}$$

Step 6: Adjusting for the transform matrix update Add and subtract $\Lambda^{(k+1)} \times_3 (D^{(k+1)})^T$ on the left:

$$\begin{aligned}
& \Lambda^{(k+1)} \times_3 (D^{(k+1)})^T - \Lambda^{(k+1)} \times_3 (\Delta D^{(k)})^T \\
&= \xi^{(k)} - \mu^{(k)} [\mathcal{B}^{(k+1)} \times_3 \Delta D^{(k)} + \Delta \mathcal{E}^{(k)}] \times_3 (D^{(k)})^T.
\end{aligned} \tag{F.31}$$

Rearranging yields the exact KKT form:

$$\Lambda^{(k+1)} \times_3 (D^{(k+1)})^T = \xi^{(k)} + \Psi^{(k)}, \tag{F.32}$$

where $\Psi^{(k)}$ denotes the total dual residual:

$$\Psi^{(k)} := \Lambda^{(k+1)} \times_3 (\Delta D^{(k)})^T - \mu^{(k)} [\mathcal{B}^{(k+1)} \times_3 \Delta D^{(k)} + \Delta \mathcal{E}^{(k)}] \times_3 (D^{(k)})^T. \tag{F.33}$$

Step 7: Taking the limit along the convergent subsequence Consider the convergent subsequence $\{k_j\}_{j=1}^\infty$ such that $(\mathcal{B}^{(k_j)}, D^{(k_j)}, \mathcal{E}^{(k_j)}, \Lambda^{(k_j)}) \rightarrow (\mathcal{B}^*, D^*, \mathcal{E}^*, \Lambda^*)$. By the scaled asymptotic-regularity condition (F.16), boundedness of $\{\mathcal{B}^{(k)}\}$ and $\{\Lambda^{(k)}\}$, and $\|D^{(k)}\|_2 = 1$, every term in $\Psi^{(k_j)}$ converges to zero. Consequently,

$$\|\Psi^{(k_j)}\|_F \rightarrow 0. \quad (\text{F.34})$$

Taking the limit as $j \rightarrow \infty$ in Eq. (F.32):

- Left-hand side: $\Lambda^{(k_j+1)} \times_3 (D^{(k_j+1)})^T \rightarrow \Lambda^* \times_3 (D^*)^T$.
- Residual: $\Psi^{(k_j)} \rightarrow 0$.
- Subgradient: By the outer semi-continuity of the limiting subdifferential operator $\partial \text{Sep}(\cdot)$, the limit of $\xi^{(k_j)}$ belongs to $\partial \text{Sep}(\mathcal{B}^*, \Omega, \vartheta)$.

Therefore, we obtain the stationarity condition for $\mathcal{B}$:

$$0 \in \partial \text{Sep}(\mathcal{B}^*, \Omega, \vartheta) - \Lambda^* \times_3 (D^*)^T. \quad (\text{F.35})$$

Step 8: Proving the continuity of Sep: Based on the structural expression of the convex-concave separation function and the Lipschitz continuity of singular value functions:

$$\begin{aligned} |\text{Sep}(\mathcal{B}^{(k+1)}, \Omega, \vartheta) - \text{Sep}(\mathcal{B}^*, \Omega, \vartheta)| &\leq \sum_{i=1}^{r_3} \vartheta_i |\text{Sep}(\tilde{\mathcal{B}}_{(i)}^{(k+1)}, \Omega) - \text{Sep}(\tilde{\mathcal{B}}_{(i)}^*, \Omega)| \\ &\leq C_{\text{Sep}} \|\mathcal{B}^{(k+1)} - \mathcal{B}^*\|_F, \end{aligned} \quad (\text{F.36})$$

where $C_{\text{Sep}} = \sqrt{r_3} \|\vartheta\|_\infty (\omega + \|\tilde{\omega}\|_1)$. Since $\mathcal{B}^{(k_j)} \rightarrow \mathcal{B}^*$, we have $\text{Sep}(\mathcal{B}^{(k_j)}, \Omega, \vartheta) \rightarrow \text{Sep}(\mathcal{B}^*, \Omega, \vartheta)$. By the outer semi-continuity of the limiting subdifferential (Lemma 15), as $k_j \rightarrow \infty$:

$$0 \in \partial \text{Sep}(\mathcal{B}^*, \Omega, \vartheta) - \Lambda^* \times_3 (D^*)^T. \quad (\text{F.37})$$

(III) Stationarity with respect to $\mathcal{E}$

From Fermat's Rule for the $\mathcal{E}$ -subproblem (F.4):

$$0 \in \partial \left(\lambda \|\mathcal{E}^{(k+1)}\|_1 + \frac{\mu^{(k)}}{2} \left\| \mathcal{E}^{(k+1)} - \left(\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} + \frac{\Lambda^{(k)}}{\mu^{(k)}} \right) \right\|_F^2 \right). \quad (\text{F.38})$$

By the sum rule:

$$0 \in \lambda \partial \|\mathcal{E}^{(k+1)}\|_1 + \mu^{(k)}(\mathcal{E}^{(k+1)} - \mathcal{Y} + \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \Lambda^{(k)} / \mu^{(k)}). \quad (\text{F.39})$$

Rearranging yields:

$$\Lambda^{(k)} + \mu^{(k)}(\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)}) \in \lambda \partial \|\mathcal{E}^{(k+1)}\|_1. \quad (\text{F.40})$$

By the multiplier update formula (F.5):

$$\Lambda^{(k+1)} \in \lambda \partial \|\mathcal{E}^{(k+1)}\|_1. \quad (\text{F.41})$$

By outer semi-continuity, as $k_j \rightarrow \infty$:

$$\Lambda^* \in \lambda \partial \|\mathcal{E}^*\|_1 \implies 0 \in \lambda \partial \|\mathcal{E}^*\|_1 - \Lambda^*. \quad (\text{F.42})$$

(IV) Primal Feasibility and Constraints

From Theorem 3:

$$\lim_{k \rightarrow \infty} \|\mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)}\|_F = 0. \quad (\text{F.43})$$

Taking the limit gives:

$$\mathcal{Y} = \mathcal{B}^* \times_3 D^* + \mathcal{E}^*. \quad (\text{F.44})$$

Since $(D^{(k)})^T D^{(k)} = I_{r_3}$ holds for all k and the column orthogonal constraint set is closed:

$$(D^*)^T D^* = I_{r_3}. \quad (\text{F.45})$$

(V) Stationarity with respect to D

Let

$$A^{(k)} := \text{unfold}_3(\mathcal{Y} - \mathcal{E}^{(k)}), \quad C^{(k+1)} := \text{unfold}_3(\mathcal{B}^{(k+1)}),$$

and

$$L^{(k)} := \text{unfold}_3(\Lambda^{(k)}).$$

The D -subproblem can be written as

$$\min_{D \in \text{St}(n_3, r_3)} \frac{\mu^{(k)}}{2} \left\| A^{(k)} + \frac{L^{(k)}}{\mu^{(k)}} - C^{(k+1)} D^T \right\|_F^2.$$

Its first-order optimality condition is

$$0 \in \mu^{(k)} \left[D^{(k+1)} (C^{(k+1)})^T C^{(k+1)} - (A^{(k)})^T C^{(k+1)} \right] - (L^{(k)})^T C^{(k+1)} + N_{\text{St}}(D^{(k+1)}). \quad (\text{F.46})$$

Let

$$\mathcal{R}^{(k+1)} := \mathcal{Y} - \mathcal{B}^{(k+1)} \times_3 D^{(k+1)} - \mathcal{E}^{(k+1)}.$$

Since

$$A^{(k)} - C^{(k+1)} (D^{(k+1)})^T = \text{unfold}_3(\mathcal{R}^{(k+1)} + \Delta \mathcal{E}^{(k)})$$

and

$$\Delta \Lambda^{(k)} = \mu^{(k)} \mathcal{R}^{(k+1)},$$

condition (F.46) becomes

$$0 \in -\text{unfold}_3(\Lambda^{(k+1)} + \mu^{(k)} \Delta \mathcal{E}^{(k)})^T C^{(k+1)} + N_{\text{St}}(D^{(k+1)}). \quad (\text{F.47})$$

Along the convergent subsequence $\{k_j\}$, condition (F.16) yields

$$\mathcal{B}^{(k_j+1)} \rightarrow \mathcal{B}^*, \quad D^{(k_j+1)} \rightarrow D^*, \quad \Lambda^{(k_j+1)} \rightarrow \Lambda^*,$$

and

$$\mu^{(k_j)} \Delta \mathcal{E}^{(k_j)} \rightarrow 0.$$

Using the closedness of the normal-cone mapping of the Stiefel manifold and passing to the limit in (F.47), we obtain

$$0 \in -\text{unfold}_3(\Lambda^*)^T \text{unfold}_3(\mathcal{B}^*) + N_{\text{St}}(D^*).$$

Therefore, $(\mathcal{B}^*, D^*, \mathcal{E}^*, \Lambda^*)$ satisfies all the generalized KKT conditions in (F.17). $\square$

Appendix G. Additional Experimental Results

This appendix reports additional real-data experiments that are omitted from the main manuscript owing to the journal length constraint. The same data preprocessing, parameter rules, initialization strategy, comparison methods, and stopping criteria reported in the main manuscript are used throughout.

Appendix G.1. Additional Hyperspectral Image Denoising Results

In addition to the Pavia Center results reported in the main manuscript, we further evaluate AC²-TRPCA on the Washington DC Mall ¹ and Pavia University (PaviaU) ² datasets under sparse-noise proportions $p \in \{0.1, 0.2, 0.3\}$. For each dataset and noise level, all methods are evaluated on the same corrupted observation.

As shown in Table G.1, AC²-TRPCA obtains the highest MPSNR at all three noise levels on the DC Mall dataset. Compared with ATNN, the corresponding gains are 5.564, 4.521, and 1.813 dB. The proposed method also yields the lowest ERGAS for $p = 0.1$ and $p = 0.2$. At the most severe corruption level $p = 0.3$, CTV achieves higher MSSIM and lower ERGAS, whereas AC²-TRPCA retains the highest MPSNR. This result indicates that the advantage of the proposed method under severe corruption is more pronounced in pixel-wise reconstruction fidelity than in structural similarity or normalized band-wise error. The visual comparison is presented in Fig. G.1.

Table G.2 provides an additional quantitative comparison on PaviaU. AC²-TRPCA again achieves the highest MPSNR and the lowest ERGAS at all three noise levels, with MPSNR improvements of 1.167, 2.235, and 1.581 dB over ATNN, respectively. CTV obtains the highest MSSIM, showing that explicit spatial smoothness remains advantageous for this structural metric. In terms of efficiency, the proposed reduced-core formulation uses approximately 31% and 40% of the average runtime required by full-size CCSVS on DC Mall and PaviaU, respectively, and is substantially faster than TCTV. These results provide additional evidence for the accuracy–efficiency trade-off of the proposed method across different hyperspectral scenes.

Appendix G.2. Additional Video Background Modeling Results

In addition to the Lobby results reported in the main manuscript, we evaluate AC²-TRPCA on the Curtain video from the I2R dataset [12]. Mixed noise is generated by combining Gaussian noise of intensity $G = 0.1$ with sparse corruption proportions $S \in \{0.1, 0.2, 0.3\}$. For each noise setting, all methods are evaluated on the same corrupted observation.

¹<https://engineering.purdue.edu/~biehl/MultiSpec/hyperspectral.html>

²https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

Table G.1: Quantitative comparison on the Washington DC Mall dataset under sparse noise. The best restoration result in each column is highlighted in bold; the noisy observation is included only as a reference.

Method	Sparse ($p = 0.1$)				Sparse ($p = 0.2$)				Sparse ($p = 0.3$)			
	MPSNR $\uparrow$	MSSIM $\uparrow$	ERGAS $\downarrow$	Time (s) $\downarrow$	MPSNR $\uparrow$	MSSIM $\uparrow$	ERGAS $\downarrow$	Time (s) $\downarrow$	MPSNR $\uparrow$	MSSIM $\uparrow$	ERGAS $\downarrow$	Time (s) $\downarrow$
Noisy	14.469	0.3470	731.046	0	11.464	0.1879	1032.342	0	9.699	0.1193	1265.520	0
ATNN	41.962	0.9963	33.156	47.0	42.419	0.9967	31.503	42.2	42.506	0.9967	31.072	55.1
CCSVS	41.029	0.9879	49.324	104.1	39.598	0.9850	54.945	145.3	37.977	0.9808	62.152	153.2
CTV	46.363	0.9986	21.598	107.0	45.124	0.9982	24.703	98.7	43.681	0.9975	28.760	125.6
E3DTV	38.736	0.9908	45.210	35.3	37.729	0.9889	50.072	41.4	36.461	0.9847	57.890	40.4
KBR	34.814	0.9781	71.013	199.4	33.825	0.9745	78.085	174.0	31.483	0.9515	99.246	218.7
LRTV	35.915	0.9783	58.954	34.2	33.900	0.9657	74.096	41.4	31.881	0.9430	94.888	40.4
TCTV	47.510	0.9954	30.422	500.4	45.708	0.9944	33.549	496.8	43.617	0.9928	37.751	517.4
Ours	47.526	0.9984	17.070	37.0	46.940	0.9981	18.200	44.2	44.319	0.9921	64.098	44.8

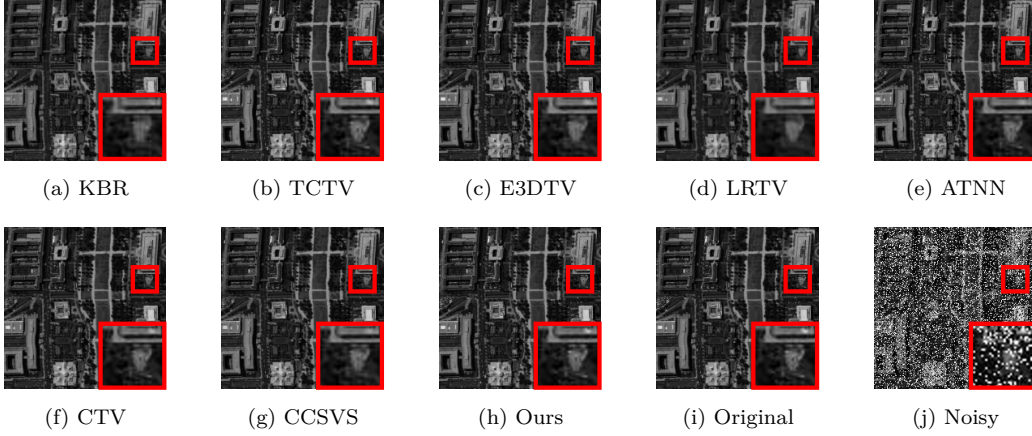


Figure G.1: Visual comparison of different denoising methods on the Washington DC Mall dataset. Bands 49, 27, and 7 are displayed as the R, G, and B channels, respectively.

Table G.2: Quantitative comparison on the PaviaU dataset under sparse noise. The best restoration result in each column is highlighted in bold; the noisy observation is included only as a reference.

Method	Sparse ($p = 0.1$)				Sparse ($p = 0.2$)				Sparse ($p = 0.3$)			
	MPSNR $\uparrow$	MSSIM $\uparrow$	ERGAS $\downarrow$	Time (s) $\downarrow$	MPSNR $\uparrow$	MSSIM $\uparrow$	ERGAS $\downarrow$	Time (s) $\downarrow$	MPSNR $\uparrow$	MSSIM $\uparrow$	ERGAS $\downarrow$	Time (s) $\downarrow$
Noisy	14.526	0.2397	695.645	0.0	11.534	0.1154	981.427	0.0	9.764	0.0696	1203.881	0.0
ATNN	46.844	0.9967	18.044	47.3	45.480	0.9960	21.059	38.5	43.801	0.9949	25.272	49.0
CCSVS	47.346	0.9842	55.724	73.2	45.239	0.9811	60.037	58.3	42.549	0.9759	65.921	72.1
CTV	47.541	0.9973	16.785	62.0	46.365	0.9967	19.138	53.9	44.928	0.9959	22.354	67.2
E3DTV	41.645	0.9895	33.341	22.0	40.312	0.9870	38.941	19.9	38.354	0.9826	47.726	22.6
KBR	36.014	0.9764	60.664	102.0	34.633	0.9664	69.554	91.5	33.256	0.9570	82.496	91.0
LRTV	39.061	0.9839	40.103	21.6	37.226	0.9746	50.052	18.2	35.085	0.9533	67.459	23.1
TCTV	46.239	0.9896	41.715	298.8	44.688	0.9878	44.833	245.3	42.993	0.9851	48.946	275.6
Ours	48.011	0.9959	15.382	30.2	47.715	0.9957	15.750	25.5	45.382	0.9938	22.108	26.8

As shown in Table G.3, AC²-TRPCA achieves the highest MPSNR under all three noise settings. Compared with ATNN, the corresponding gains are 1.299, 1.834, and 2.372 dB. Relative to the second-best method, the gains are 0.347, 0.729, and 0.921 dB, respectively. The proposed method also records the shortest runtime in every setting. E3DTV obtains the highest MSSIM, which may be attributed to its explicit 3-D total-variation prior that directly promotes local spatiotemporal smoothness and structural continuity. Overall, the Curtain results confirm the favorable reconstruction-accuracy and computational-efficiency trade-off of the proposed reduced-core formulation. The visual comparison is shown in Fig. G.2.

Table G.3: Quantitative comparison on the Curtain video under mixed Gaussian and sparse noise. The best restoration result in each column is highlighted in bold; the noisy observation is included only as a reference.

Method	$G = 0.1, S = 0.1$			$G = 0.1, S = 0.2$			$G = 0.1, S = 0.3$		
	MPSNR $\uparrow$	MSSIM $\uparrow$	Time (s) $\downarrow$	MPSNR $\uparrow$	MSSIM $\uparrow$	Time (s) $\downarrow$	MPSNR $\uparrow$	MSSIM $\uparrow$	Time (s) $\downarrow$
Noisy	14.248	0.1153	0	11.852	0.0671	0	10.309	0.0444	0
ATNN	31.994	0.7935	20.4	31.110	0.7575	15.4	30.064	0.7116	16.8
CCSVS	25.873	0.4957	34.2	24.612	0.4340	25.8	23.067	0.3633	26.4
CTV	32.248	0.8603	27.2	31.711	0.8450	28.5	31.057	0.8247	27.6
E3DTV	32.625	0.9023	9.1	32.051	0.8950	9.0	31.294	0.8865	10.1
KBR	31.445	0.8574	50.7	31.101	0.8480	53.6	30.590	0.8364	49.3
LRTV	32.946	0.8665	10.4	32.215	0.8482	8.3	31.515	0.8321	9.9
TCTV	29.876	0.7109	109.7	28.891	0.6666	131.2	27.797	0.6147	121.2
Ours	33.293	0.8823	9.0	32.944	0.8724	7.2	32.436	0.8589	7.5

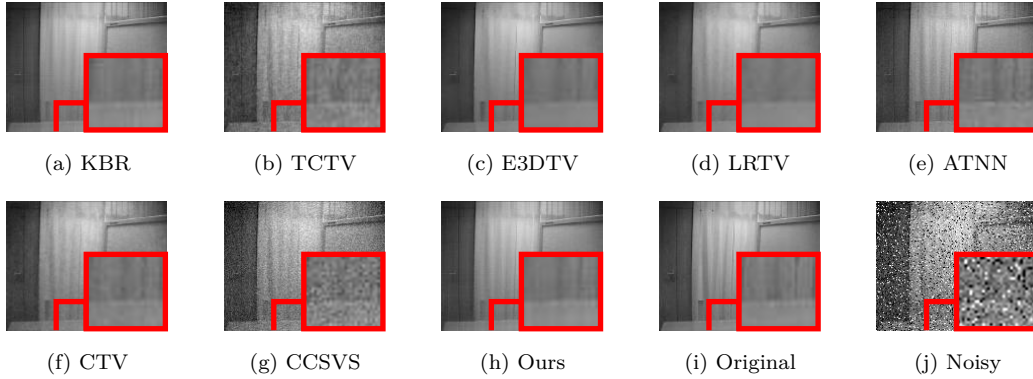


Figure G.2: Visual comparison of different background-recovery methods on the Curtain video under mixed Gaussian and sparse noise.

Appendix G.3. Additional Phase-Transition Analysis

We further investigate the empirical recovery boundary of AC²-TRPCA through phase-transition experiments. The tensor size is fixed at $n_1 = n_2 = n_3 = 100$. We vary the tubal-rank ratio $\rho_r := R/n$ and the sparse-corruption ratio $\rho_s := \|\mathcal{E}_0\|_0/n^3$ over a 25×25 grid. At each grid point, five independent Monte Carlo trials are performed. A trial is declared successful when the relative low-rank recovery error $\text{RelErr}_{\mathcal{T}} = \|\hat{\mathcal{T}} - \mathcal{T}_0\|_F / \|\mathcal{T}_0\|_F$ is below 10^{-3} . The displayed value is the empirical success rate over the five trials.

Figure G.3 presents the phase-transition diagrams for $r_3 \in \{10, 15, 20\}$. In all three cases, the recovery probability decreases as either the tubal-rank ratio or the corruption ratio increases, producing a clear transition between successful and unsuccessful recovery. The successful region is broadest for $r_3 = 10$ and contracts moderately as r_3 increases to 15 and 20. Under the present synthetic model, this behavior suggests that a more compact prescribed reduced dimension can facilitate recovery by reducing the size of the core representation. Nevertheless, the transition remains well defined throughout the tested range, indicating moderate robustness to the choice of r_3 . These empirical observations are consistent with the role of the prescribed dimension r_3 in the sufficient sparsity condition of Theorem 1 and provide an empirical view of how the recovery region changes with the size of the reduced core.

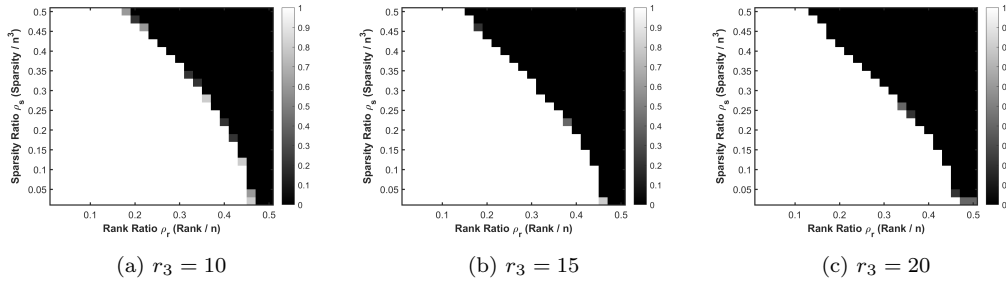


Figure G.3: Empirical phase-transition diagrams of AC²-TRPCA under different prescribed reduced dimensions r_3 . White denotes an empirical success rate of 1, black denotes a success rate of 0, and intermediate gray levels represent success rates between 0 and 1.

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